PHYSICS

1) Two point charges Q and – 3Q are placed at some distance apart. If the electric field at the location of Q is $\vec{E}$ then electric field at location of – 3Q is :

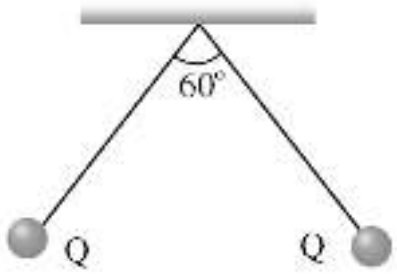
(1) $\vec{E}$

(2) $\vec{E}/3$

(3) $-3\vec{E}$

(4) $-\vec{E}/3$

2) Two small spherical balls each carrying a charge Q = 10 μC are suspended by two insulating threads of equal lengths 1m each, from a point fixed in the ceiling. It is found that in equilibrium threads are separated by an angle 60° between them, as shown in the figure. What is the tension in the threads (Given: $\frac{1}{(4\pi\varepsilon_0)} = 9 \times 10^9 \text{Nm/C}^2$)



(1) 18 N

(2) 1.8 N

(3) 0.18 N

(4) None of the above

3) Two point charges +8q and –2q are located at x = 0 and x = L respectively. The location of a point on the x-axis at which the net electric field due to these two point charges is zero is

(1) 8 L

(2) 4 L

(3) 2 L

(4) $\frac{L}{4}$

4) Four charges equal to – Q are placed at the four corners of a square and a charge q is at its centre. If the system is in equilibrium the value of q is

(1) $-\frac{Q}{4}(1+2\sqrt{2})$

(2) $\frac{Q}{4}(1+2\sqrt{2})$

(3) $-\frac{Q}{2}(1+2\sqrt{2})$

(4) $\frac{Q}{2}(1+2\sqrt{2})$

5) Two spheres A and B of radius 4 cm and 6 cm are given charges of 80 μC and 40 μC respectively. If they are connected by a fine wire, the amount of charge flowing from one to the other is

(1) $20\mu C$ from A to B

(2) $16\mu C$ from A to B

(3) $32\mu C$ from B to A

(4) $32\mu C$ from A to B

6) Two equally charged, identical metal spheres A and B repel each other with a force 'F'. The spheres are kept fixed with a distance 'r' between them. A third identical, but uncharged sphere C is brought in contact with A and then placed at the mid-point of the line joining A and B. The magnitude of the net electric force on C is

(1) F

(2) 3F/4

(3) F/2

(4) F/4

7) **Assertion :** Mass of ion is slightly differed from its element.
Reason: Ion is formed, when some electrons are removed or added so mass changes.

(1) Both Assertion and Reason are correct and Reason is correct Explanation of Assertion.

(2) Both assertion and Reason are correct but Reason is not correct explanation of Assertion.

(3) Assertion is correct, Reason is wrong.

(4) Both Assertion and Reason are wrong.

8) **Assertion :** Charge is quantized
Reason : Charge, which is less than 1 C is not possible

(1) Both Assertion and Reason are correct and Reason is correct Explanation of Assertion.

(2) Both assertion and Reason are correct but Reason is not correct explanation of Assertion.

(3) Assertion is correct, Reason is wrong.

(4) Both Assertion and Reason are wrong.

9) Match the column:

	Column-I		**Column-II**
(A)	$y = \sin x - \cos x$ $\frac{dy}{dx}$ at $x = \frac{\pi}{2}$	(P)	2

(B)	$y = e^{7x}$ $\frac{dy}{dx}$ at x = 0	(Q)	1
(C)	$y = \log_e x$ $\frac{dy}{dx}$ at x = 1	(R)	7
(D)	$y = x$ $\frac{dy}{dx}$ at x = 2	(S)	Zero
		(T)	None

(1) A → Q; B → R; C → Q; D → Q;

(2) A → Q; B → P; C → S; D → T;

(3) A → S; B → Q; C → R; D → S;

(4) A → P; B → R; C → Q; D → S;

10) Match the column:

	Column - I		**Column - II**
(A)	$\int_0^2 5\,dx$	(P)	1
(B)	$\int_1^2 x\,dx$	(Q)	10
(C)	$\int_1^2 x^{-1}\,dx$	(R)	1.5
(D)	$\int_0^\infty e^{-x}\,dx$	(S)	$\log_e 2$

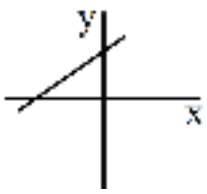
(1) A → P; B → Q; C → R; D → S

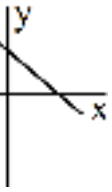
(2) A → R; B → S; C → P; D → Q

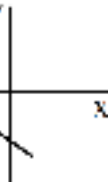
(3) A → Q; B → R; C → S; D → P

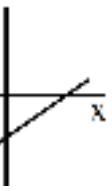
(4) A → S; B → P; C → Q; D → R

11) Curve represented by equation y = 3x + 4 is -

(1) 

(2) 

(3) 

(4) 

12) tan 210° is equal to -

(1) $\frac{1}{\sqrt{3}}$

(2) $-\frac{1}{\sqrt{3}}$

(3) $\sqrt{3}$

(4) $\bar{\sqrt{3}}$

13) Two identical charged particles of same mass are placed at such a distance that the coulomb's force of repulsion is equal to the gravitational force of attraction. Then the specific charge of these particles is :

(1) $\sqrt{g}$

(2) $\sqrt{4\pi\in_0}$

(3) $\sqrt{4\pi\in_0 G}$

(4) $\sqrt{\frac{G}{4\pi\in_0}}$

14) Electric charges of 1μC, –1μC and 2μC are placed in air at the corners A, B and C respectively of an equilateral triangle ABC having length of each side 10 cm. The resultant force on the charge at C is :-

(1) 0.9 N

(2) 1.8 N

(3) 2.7 N

(4) 3.6 N

15) ABC is a right angled triangle in which AB = 3 cm and BC = 4cm and ∠ ABC = π/2. The three charges +15 esu, +12 esu and – 20 e.s.u are placed respectively on A, B and C. The force acting on B is :-

(1) 125 dynes

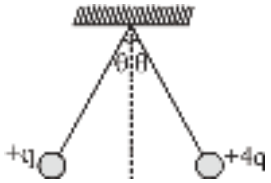
(2) 35 dynes

(3) 25 dynes

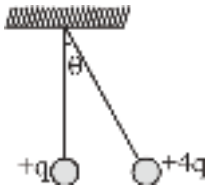
(4) Zero

16) Two metal spheres of same mass are suspended from a common point by a light insulating string. The length of each string is the same. The spheres are given electric charges +q on one and +4q on the other. Which of the following diagrams best shows the resulting position of spheres ?

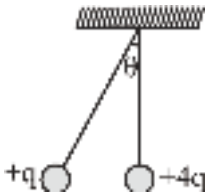
(1)



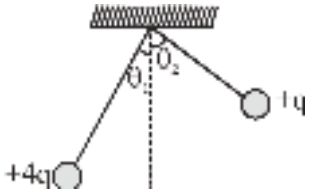
(2)



(3)



(4)



17) There are 2 neutral bodies A & B. Both are rubbed with each other so that A acquires positive charge and B acquires negative charge, then body having high work function is :-

(1) A

(2) B

(3) Both have same work function

(4) Can't say

18) A charge water drop of radius 'r' is in equilibrium in an electric field. If charge on it is equal to charge of an electron, then intensity of electric field will be : (Density of water = ρ, assume gravity)

(1) $\frac{8\pi r^3\rho}{3e}g$

(2) $\frac{\pi r^3\rho}{3e}g$

(3) $\frac{6\pi r^3\rho}{15e}g$

(4) $\frac{4\pi r^3\rho}{3e}g$

19) Surface charge density of a thin disc having radius R varies with distance r from centre as $\sigma = \frac{\sigma_0 r^2}{R^2}$, ($r \neq 0$) then total charge of disc is :-

(1) $2\pi\sigma_0 R^2$

(2) $\sqrt{2}\pi\sigma_0 R^2$

(3) $2\sqrt{2}\pi\sigma_0 R^2$

(4) $\frac{\pi\sigma_0 R^2}{2}$

20) A uniformly charged ring of radius R has charge Q. Electric field at axis of ring at distance $\sqrt{3}R$ from its centre is :-

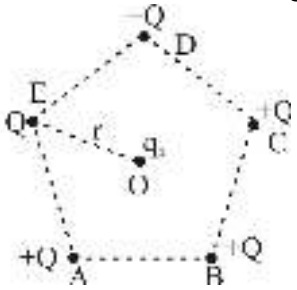
(1) $\sqrt{3}\frac{kQ}{R^2}$

(2) $\frac{2}{3\sqrt{3}}\frac{kQ}{R^2}$

(3) $\frac{\sqrt{3}}{8}\frac{kQ}{R^2}$

(4) $\frac{\sqrt{3}}{16}\frac{kQ}{R^2}$

21) Five charges of equal values placed at corners of a regular pentagon of side 'a'. The force on q which is placed at centroid 'O' is :-



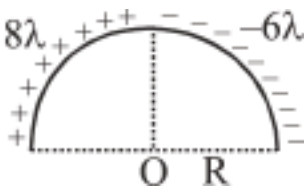
(1) zero

(2) $\frac{2kQq_0}{r^2}$, along OE

(3) $\frac{2kQq_0}{r^2}$, along EO

(4) $\frac{kQq_0}{r^2}$, along OE

22) Electric field intensity at centre of semicircular arc as shown in fig is :-



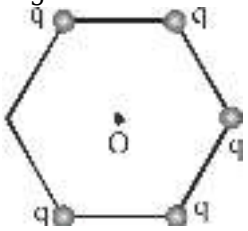
(1) $10\frac{k\lambda}{R}$

(2) $10\sqrt{2}\frac{k\lambda}{R}$

(3) $5\frac{k\lambda}{R}$

(4) $5\sqrt{2}\frac{k\lambda}{R}$

23) Five point charges each having magnitude q' are placed at the corners of regular hexagon as shown in figure. Net electric field at the centre 'O' is $\vec{E}$. To get net electric field at 'O' to be $6\vec{E}$, charge placed on the remaining sixth corner should be :-



(1) 6 q

(2) – 6 q

(3) 5 q

(4) – 5 q

24) The vector $\vec{P}$ makes 120° with the x-axis and the vector Q makes 30° with the y-axis, what is their resultant :

(1) P + Q

(2) P – Q

(3) $\sqrt{P^2+Q^2}$

(4) $\sqrt{P^2-Q^2}$

25) If $\vec{A}$ = 5 cos θ $\hat{i}-3\sin\theta\hat{j}$ then which vector will be perpendicular to $\vec{A}$:

(1) $5\sin\theta\hat{i}+3\cos\theta\hat{j}$

(2) $3\sin\theta\hat{i}-5\cos\theta\hat{j}$

(3) $3\sin\theta\hat{i}+5\cos\theta\hat{j}$

(4) $5\sin\theta\hat{i}-3\cos\theta\hat{j}$

26) Which of the following set of forces can produce zero resultant :-

(1) 20 N, 30 N, & 60 N

(2) 20 N, 40 N, & 60 N

(3) 10 N, 20 N & 40 N

(4) 10 N, 10 N & 30 N

27) Two forces, each of magnitude F have a resultant of the same magnitude F. The angle between the two forces is :-

(1) 45°

(2) 120°

(3) 150°

(4) 60°

28) If $\hat{a}$ and $\hat{b}$ are non-collinear unit vectors and $|\hat{a}+\hat{b}|=\sqrt{3}$ then the value of $(2\hat{a}-5\hat{b}).(3\hat{a}+\hat{b})$ is :

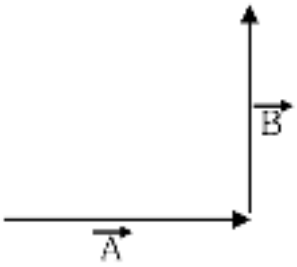
(1) $\frac{41}{2}$

(2) $\frac{11}{2}$

(3) $-\frac{11}{2}$

(4) $-\frac{41}{2}$

29) Vectors $\vec{A}$ and $\vec{B}$ are shown in figure then diagram of $\vec{B}-\vec{A}$ is



(1) 

(2) 

(3) 

(4) None of these

30) **Assertion (A) :-** For two vectors $\vec{a}$ & $\vec{b}$, $\vec{a}\times\vec{b}=\vec{a}.\vec{b}$ for $\theta = 45°$.
Reason (R) :- Two vectors are equal when their magnitude and direction both are same.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

31) With respect to a rectangular cartesian co-ordinate system three vectors are expressed as $\vec{a}=4\hat{i}-\hat{j}, \vec{b}=-3\hat{i}+2\hat{j}, \ \vec{c}=-\hat{k}$ where $\hat{i}, \hat{j}$ and $\hat{k}$ are unit vectors along x, y, z axis respectively. The unit vector along the direction of sum of these vectors is :

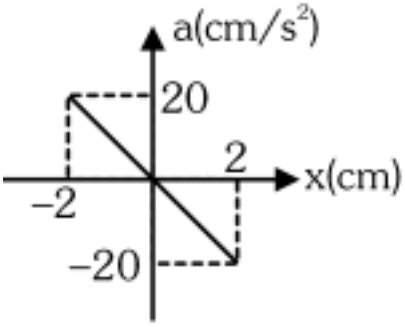
(1) $\hat{r} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} - \hat{k})$

(2) $\hat{r} = \frac{1}{\sqrt{2}}(\hat{i} + \hat{j} - \hat{k})$

(3) $\hat{r} = \frac{1}{3}(\hat{i} - \hat{j} + \hat{k})$

(4) $\hat{r} = \frac{1}{\sqrt{3}}(\hat{i} + \hat{j} + \hat{k})$

32) The acceleration-displacement graph of a particle executing simple harmonic motion is shown in the figure. The frequency of oscillation is



(1) $\frac{\sqrt{2.5}}{\pi} s^{-1}$

(2) $2\pi\sqrt{10} s^{-1}$

(3) $\frac{1}{2\pi}\sqrt{5} s^{-1}$

(4) $\frac{1}{2\pi}\sqrt{20} s^{-1}$

33) A particle at end of a spring executes simple harmonic motion with a period t_1, while the corresponding period of another spring is t_2. If two springs are connected in series then time period is T, them which one is true? (Consider same mass of particle in all cases)

(1) $T = t_1 + t_2$

(2) $T^2 = t_1^2 + t_2^2$

(3) $T = t_1^{-1} + t_2^{-1}$

(4) $T = t_1^{-2} + t_2^{-2}$

34) A particle is vibrating in simple harmonic motion with amplitude of 4 cm. The displacement of the particle when kinetic energy becomes half of total energy is

(1) 10 cm

(2) $\sqrt{2}$ cm

(3) 2 cm

(4) $2\sqrt{2}$ cm

35) A mass M is suspended from a spring of negligible mass. The spring is pulled a little and then released so that the mass executes SHM of time period T. If the mass is increased by m, the time period becomes 5T/3. Then the ratio of m/M is

(1) 3/5

(2) 25/9

(3) 16/9

(4) 5/3

36) The equation of simple harmonic motion of a particle is $a + 4\pi^2 x = 0$ where a the instantaneous linear acceleration at displacement x. The frequency of motion is

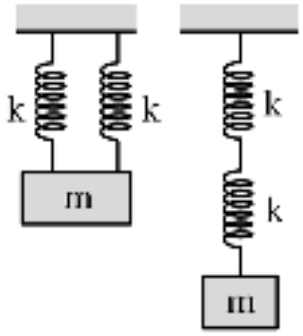
(1) 1 Hz

(2) 4π Hz

(3) $\frac{1}{4}$Hz

(4) 4 Hz

37) Two identical springs are connected in parallel and series as shown in figure. If f_p and f_s are



frequencies of arrangement, what is f_s / f_p ?

(1) 1 : 2

(2) 2 : 1

(3) 1 : 3

(4) 3 : 1

38) The displacement of an object attached to a spring and executing simple harmonic motion is given by $X = 2 \times 10^{-2} (\cos \pi t)$. The time at which the maximum speed first occurs is

(1) 0.25 s

(2) 0.75 s

(3) 0.125 s

(4) 0.5 s

39) **Assertion (A) :-** The graph of total mechanical energy of a particle in SHM w.r.t. position is a straight line with zero slope.
Reason (R) :- Total energy of particle in SHM remains constant throughout its motion.

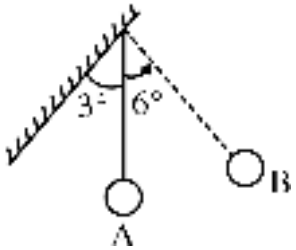
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true.

40) A pendulum of length 10 cm is hanged by a wall making an angle 3° with vertical. It is swung to position B. Time period of pendulum will be (Assuming every collision to be elastic)



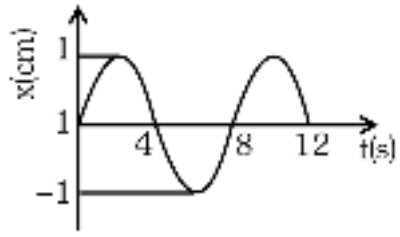
(1) $\frac{\pi}{5}s$

(2) $\frac{2\pi}{15}s$

(3) $\frac{\pi}{6}s$

(4) subsequent motion will not be periodic

41) The x-t graph of a particle undergoing simple harmonic motion is shown below. The acceleration of the particle at $t = \frac{2}{3}s$



(1) $\frac{\sqrt{3}}{32}\pi^2\frac{cm}{s^2}$

(2) $-\frac{\pi^2}{32}\frac{cm}{s^2}$

(3) $\frac{\pi^2}{32}\frac{cm}{s^2}$

(4) $-\frac{\sqrt{3}}{32}\pi^2\frac{cm}{s^2}$

42) The equation of a wave travelling on a string is

$y = 4\sin\frac{\pi}{2}\left(8t - \frac{x}{8}\right)$

if x and y are in cm, then velocity of the wave is

(1) 64 cm s^{-1} in –x direction

(2) 32 cm s^{-1} in –x direction

(3) 32 cm s^{-1} in +x direction

(4) 64 cm s^{-1} in +x direction

43) A wave of frequency 500 Hz has a velocity of 350 m/s. The approx distance between two nearest point on the wave which are at 60° phase difference is :-

(1) 70 cm

(2) 0.7 m

(3) 12.0 cm

(4) 120.0

44) When a sound wave of frequency 300 Hz passes through a medium, the maximum displacement of a particle of the medium is 0.1 cm, the maximum velocity of the particle is equal to :

(1) 60 π cm s^{-1}

(2) 30 π cm s^{-1}

(3) 30 cm s^{-1}

(4) 60 cm s^{-1}

45) A simple pendulum of length L has mass M and it oscillates freely with amplitude A. At the extreme position. It's potential energy is (g = acceleration due to gravity)

(1) $\frac{MgA^2}{2L}$

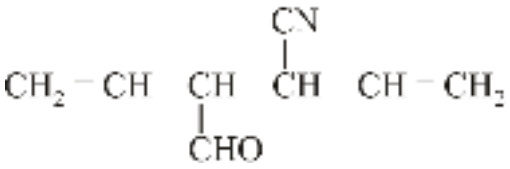
(2) $\frac{MgA}{2L}$

(3) $\frac{MgA^2}{L}$

(4) $\frac{2MgA^2}{L}$

CHEMISTRY

1) The correct IUPAC name of the following?



(1) 3-cyano-2-ethenyl pent-4-enal

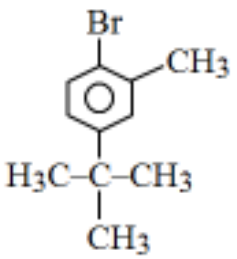
(2) 2-ethenyl-3-formyl pent-4-ene nitrile

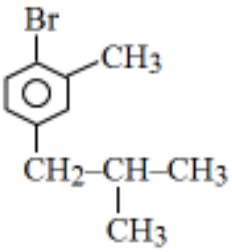
(3) 2,3-diethenyl-4-oxobutane nitrile

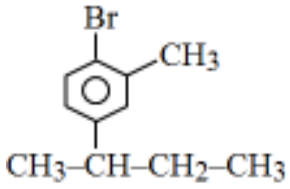
(4) 3-cyano-4-formyl hexa-1,5-diene

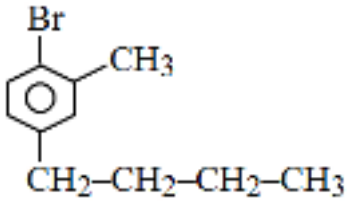
2) 1-Bromo-4-sec. butyl-2-methyl benzene is :-

(1)

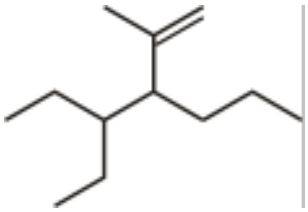


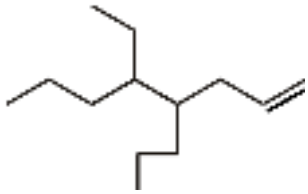
(2) 

(3) 

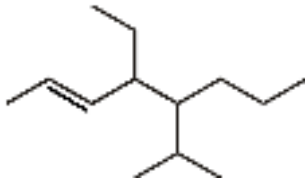
(4) 

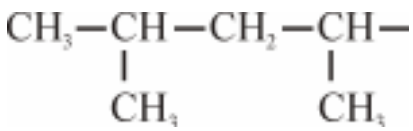
3) Which of the following is correct structure of 4-ethyl-5-isopropyl oct-2-enc.

(1) 

(2) 

(3) 

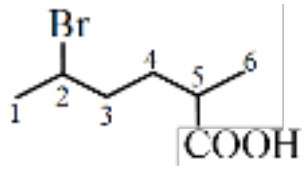
(4) 

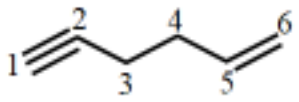
4) 

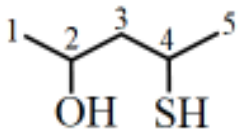
Radical has IUPAC name–

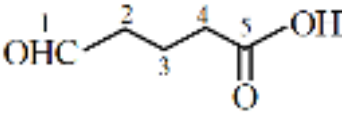
(1) 4-methyl pentyl
(2) 1,3-Dimethyl butyl
(3) 1,4-Dimethyl butyl
(4) 3-methyl pentyl

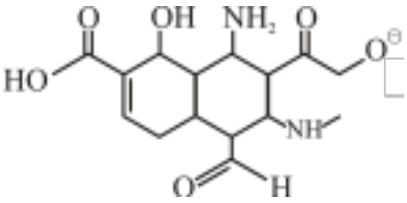
5) Which of the following has correct IUPAC numbering ?

(1) 

(2) 

(3) 

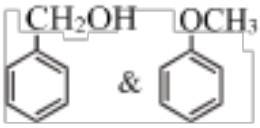
(4) 



6) How many functional group are present in following compound :-

(1) 7
(2) 6
(3) 8
(4) 9

7) Identify pair of homologues :-

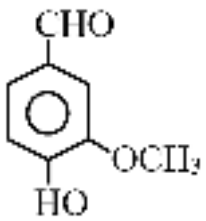
(1) 

(2) 

(3) 

(4) $CH_3–CH_2–NH_2$ & $(CH_3)_3N$

8)

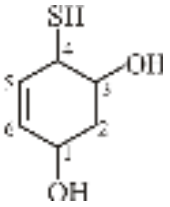
IUPAC name of 

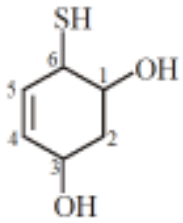
is

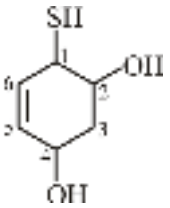
(1) 4-Formyl-2-Hydroxy anisole
(2) 4-Hydroxy-3-methoxy benzaldehyde
(3) 4-Formyl-2-methoxy phenol

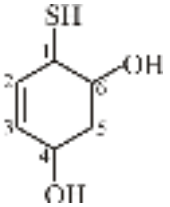
(4) All are wrong

9) Select the structure with correct numbering for the IUPAC name of the given compound?

(1) 

(2) 

(3) 

(4) 

10)

Minimum carbon required to show chain isomerism in ketone

(1) 5
(2) 3
(3) 7
(4) 6

11) Match the columns :-

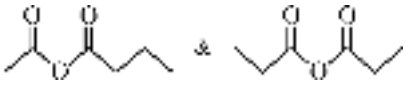
	Column-I		**Column-II**
(A)	CH_3–OH, CH_3–CH_2–OH	(i)	Chain isomer
(B)	CH_3–CH_2–NH_2, CH_3–NH–CH_3	(ii)	Ring chain isomer
(C)	CH_3–CH=CH_2, 	(iii)	Functional isomer
(D)	CH_3–CH_2–CH_2–CH_3 CH_3–CH(–CH_3) CH_3	(iv)	Homologous

(1) A-(i), B-(ii), C-(iii), D-(iv)

(2) A-(ii), B-(iii), C-(iv), D-(i)

(3) A-(iv), B-(iii), C-(ii), D-(i)

(4) A-(iii), B-(i), C-(iv), D-(ii)

12)  are :

(1) Metamers

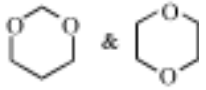
(2) Chain isomers

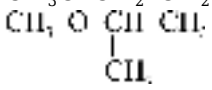
(3) Identical

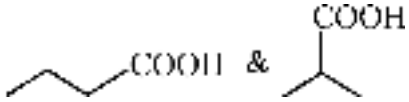
(4) Functional isomers

13) Which of following pairs of compounds are not position isomers ?

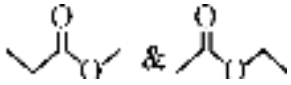
(1) 

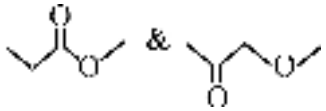
(2) 

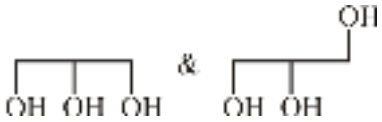
(3) $CH_3O–CH_2–CH_2–CH_3$ &


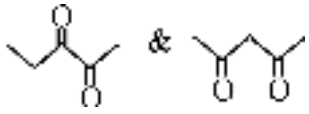
(4) 

14) Which of the following pairs of compounds are functional group isomers :-

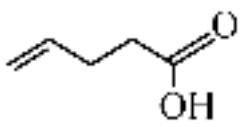
(1) 

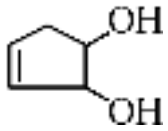
(2) 

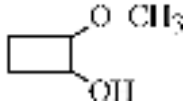
(3) 

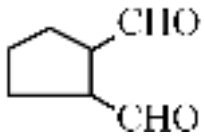
(4) 

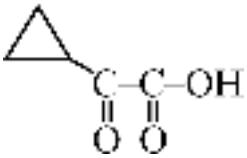
15)

Which of the following is/are isomer of 

(1) 

(2) 

(3) 

(4) 

16) Which of the following homologues series general formula is not correctly matched.

(1) Anhydride → $C_nH_{2n-4}O_3$

(2) Aldehyde → $C_nH_{2n}O$

(3) Alkyne → C_nH_{2n-2}

(4) Amide → C_nH_{2n+1} NO

17) **Statement-I :** Acetic acid was synthesised by berthelot.
Statement-II : First organic compound synthesised in lab was urea.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Statement-I correct, Statement-II incorrect

(4) Statement-I incorrect, Statement-II correct

18) **Statement-I :** Ethene has 116.6 and 121.7 bond angle respectively.
Statement-II : First member of alkene is : CH_2 but has a very short life.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Statement-I correct, Statement-II incorrect

(4) Statement-I incorrect, Statement-II correct

19) Number of σ bond present in mesityl oxide is :

(1) 16

(2) 18

(3) 20

(4) 22

20) **Statement-I :** Pyran is heterocyclic and alicyclic compound.
Statement-II : Furan is hetero cyclic and aromatic compound.

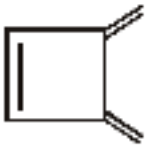
(1) Statement-I & statement-II both are correct

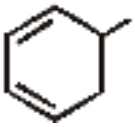
(2) Statement-I is correct but statement-II is incorrect

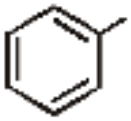
(3) Statement-I is incorrect but statement-II is correct

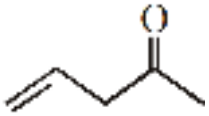
(4) Statement-I & statement-II both are incorrect

21) Which of the following molecules have only sp^2 hybridised carbon atoms ?

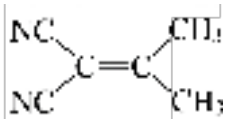
(1) 

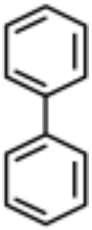
(2) 

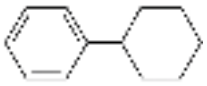
(3) 

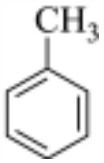
(4) 

22) In which of the following molecules all atoms are co-planar.

(1) 

(2) 

(3) 

(4) 

23) Which of the following rate law expression represent zero order reaction :-

(1) $K(A)^{3/2}\ (B)^{-1}\ (C)^{1/2}$

(2) $K(A)^{0}\ (B)^{2}$

(3) $K(A)^{1/2}\ (B)^{1/2}\ (C)^{-1}$

(4) $K(A)^{2}\ (B)^{1}$

24) For a reaction A + B → C + D, the rate constant $2.5 \times 10^{2}\ s^{-1}$. If the concentration of reactants

are reduced to half, then the value of rate constant is :

(1) $2.5 \times 10^2\ s^{-1}$

(2) $1.25 \times 10^2\ s^{-1}$

(3) $625 \times 10^2\ s^{-1}$

(4) none of these

25) In the following reaction : xA → yB

$$\log\left[-\frac{d[A]}{dt}\right] = \log\left[\frac{d[B]}{dt}\right] + \log 2$$

where –ve sign indicates rate of disappearance of the reactant. Thus, x : y is :

(1) 1 : 2

(2) 2 : 1

(3) 3 : 1

(4) 3 : 10

26) For the reaction

$2SO_{2(g)} + O_{2(g)} \xrightarrow{Pt} 2SO_{3(g)}$,

$r = K[SO_2]\,[SO_3]^{-1/2}$

which of the following is incorrect ?

(1) The overall order of reaction is +1/2

(2) The reaction slows down as the product SO_3 builds up

(3) The order of reaction with respect to SO_2 is 1

(4) Unit of rate constant is L/mol-sec

27) For the reaction 2A + B → C, following results are obtained-

$[A]_0$	$[B]_0$	Initial Rate of reaction (in M sec^{-1})
0.1 M	0.1 M	6×10^{-4}
0.4 M	0.1 M	12×10^{-4}
0.4 M	0.2 M	48×10^{-4}

then correct rate equation will be :-

(1) $r = k[A]^2\,[B]^2$

(2) $r = k[A]^{1/2}\,[B]^2$

(3) $r = k[A]^1\,[B]^2$

(4) $r = k[A]^2\,[B]^1$

28) The rate equation of a reaction is as follows :

$r = k\,[A]^{0.5}[B]\,[C]^{0.5}$

Which statement about the above equation is wrong ?

(1) Order with respect to each A and C is 0.5

(2) Unit of specific reaction rate is $mol^{-1}L\ s^{-1}$

(3) Order with respect to B is one

(4) Total order of the reaction is 2.5

29) A gaseous reaction, $2A(g) \rightarrow B(g) + 5C(g)$ shows increase in pressure from 80 mm to 100 mm in 5 minutes. The rate of disappearance of A is :-

(1) 4 mm min^{-1}

(2) 8 mm min^{-1}

(3) 16 mm min^{-1}

(4) 2 mm min^{-1}

30)

For the reaction
$2H_2 + 2NO \rightarrow N_2 + 2H_2O$
the following mechanism has been proposed :-
(i) $2NO \rightleftharpoons N_2O_2$ (fast)

(ii) $N_2O_2 + H_2 \xrightarrow{k_2} N_2O + H_2O$ (slow) (iii) $N_2O + H_2 \rightarrow N_2 + H_2O$ (fast)
then what will be the rate law of this reaction ?

(1) $k[NO][H_2]^2$

(2) $k[NO][H_2]$

(3) $k[NO]^0[H_2]^3$

(4) $k[NO]^2[H_2]$

31) $A_2 + B_2 \rightarrow 2AB$; R.O.R = $k[A_2]^a\ [B_2]^b$

Initial$[A_2]$	Initial$[B_2]$	R.O.R(r) Ms^{-1}
0.2	0.2	0.04
0.1	0.4	0.04
0.2	0.4	0.08

Order of reaction with respect to A_2 and B_2 are respectively :

(1) a = 1, b = 1

(2) a = 2, b = 0

(3) a = 2, b = 1

(4) None

32) In a reaction, $n_1A + n_2B \rightarrow m_1C + m_2D$, 5M of reactant A are allowed to react with 3M of reactant B. After 5 second, the concentration of A was found to be 4M. Rate of decomposition A and rate of formation of D respectively :-

(1) $0.2 M sec^{-1}; \left(\frac{m_2}{m_1} \times 0.2\right) M sec^{-1}$

(2) $0.2 M sec^{-1}; \left(\frac{n_2}{m_2} \times 0.2\right) M sec^{-1}$

(3) $0.1 M sec^{-1}; \left(\frac{m_2}{n_1} \times 0.2\right) M sec^{-1}$

(4) $0.2 M sec^{-1}; \left(\frac{m_2}{n_1} \times 0.2\right) M sec^{-1}$

33) In the following reaction, $2H_2O_2 \rightarrow 2H_2O + O_2$ rate of formation of O_2 is 36 gL^{-1} min^{-1}, what is rate of disappearance of H_2O_2 in:

(1) 2.250 mol L^{-1} min^{-1}

(2) 72 mol L^{-1} min^{-1}

(3) 1.50 mol L^{-1} min^{-1}

(4) 38.25 mol L^{-1} min^{-1}

34) For a reaction $2NH_3 \rightarrow N_2 + 3H_2$. It is observed that $\frac{-d[NH_3]}{dt} = K_1[NH_3]$, $\frac{d[N_2]}{dt} = K_2[NH_3]$, $\frac{d[H_2]}{dt} = K_3[NH_3]$. What is the relation between K_1, K_2 and K_3 ?

(1) $K_1 = K_2 = K_3$

(2) $3K_1 = 6K_2 = 2K_3$

(3) $2K_1 = 3K_2 = 6K_3$

(4) $6K_1 = 3K_2 = 2K_3$

35) If the rate law of a reaction $nA \rightarrow B$ is expressed as :

$$Rate = -\frac{1}{n}\frac{d[A]}{dt} = +\frac{d[B]}{dt} = K[A]^x$$

The unit of rate constant will be -

(1) $mol^x / L^x / s$

(2) $L^x / mol^x / s$

(3) $mol^{(1-x)}.L^{(x-1)}.s^{-1}$

(4) $mol^{(x-1)}.L^{(1-x)}.s^{-1}$

36)

For a hypothetical reaction,

$A + 3B \rightarrow P \quad \Delta H = -2x$ kJ/mole of A

& $M \rightarrow 2Q + R \quad \Delta H = +x$ kJ/mole of M

If these reactions are carried simultaneously in a reactor such that temperature is not changing. If rate of disappearance of B is y M sec^{-1} then rate of formation (in M sec^{-1}) of Q is :-

(1) $\frac{2}{3}y$

(2) $\frac{3}{2}y$

(3) $\frac{4}{3}y$

(4) $\frac{3}{4}y$

37)

The following data is given for reaction between A and B :-

S.No.	[A] $mol.1^{-1}$	[B] $mol.1^{-1}$	Rate $mol.1^{-1}sec^{-1}$
I	1×10^{-2}	2×10^{-2}	2×10^{-4}
II	2×10^{-2}	2×10^{-2}	4×10^{-4}
III	2×10^{-2}	4×10^{-2}	8×10^{-4}

Which of the following are correct statements -
(a) Rate constant of the reaction 10^{-4}
(b) Rate law of the reaction is k[A][B] (c) Rate of reaction increases four times on doubling the concentration of both the reactant.

(1) a, b and c

(2) a and b

(3) b and c

(4) c alone

38)

In the reaction xA → yB,

$$0.48 + \log\left\{-\frac{d[A]}{dt}\right\} = \log\left\{+\frac{d[B]}{dt}\right\} + 0.7$$

Then, x : y is

(1) 3 : 1

(2) 3 : 5

(3) 5 : 3

(4) None

39) For the gaseous phase reaction: A(g) ⟶ product, occurring at constant volume the correct relation between $\frac{dC_A}{dt}, \frac{dn_A}{dt}$ and $\frac{dP_A}{dt}$ is:
[C_A ⟶ Concentration of reactant A,
n_A ⟶ Moles of reactant A,
P_A ⟶ Partial pressure of reactant A]

(1) $\frac{dC_A}{dt} = \frac{dn_A}{dt} = \frac{dP_A}{dt}$

(2) $\frac{dC_A}{dt} = \left(\frac{dP_A}{dt}\right) RT = \frac{1}{V}\frac{dn_A}{dt}$

(3) $\frac{dC_A}{dt} = \frac{1}{RT}\frac{dP_A}{dt} = \frac{1}{V}\frac{dn_A}{dt}$

(4) $\frac{dC_A}{dt} = \frac{1}{V}\frac{dn_A}{dt} = \frac{1}{V}\frac{dP_A}{dt}$

40) Reaction $A \xrightarrow{K=10^{-3}\,Mol^{-1}L\,Sec^{-1}} B$. Which of the following is the correct rate equation :-

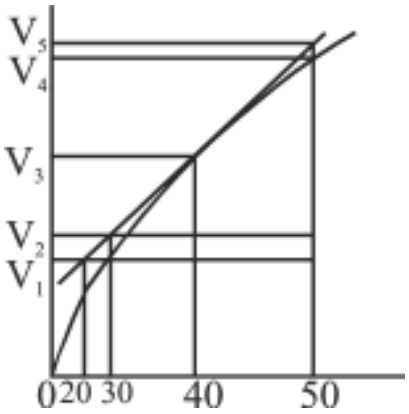
(1) $r = K[A]^2$

(2) $r = K[A]$

(3) $r = K[A]^3$

(4) None of these

41) A graph of volume of hydrogen released vs time for the reaction between zinc and dil. HCl is given in figure. On the basis of this mark the correct option.



(1) Average rate upto 40 seconds is $\frac{V_3 - V_2}{40}$

(2) Average rate upto 40 seconds is $\frac{V_3 - V_2}{40 - 30}$

(3) Average rate upto 40 seconds is $\frac{V_3}{40}$

(4) Average rate upto 40 seconds is $\frac{V_3 - V_1}{40 - 20}$

42) For hypothetical reaction A → B takes place according to

$A \underset{}{\overset{K_1}{\rightleftharpoons}} C$; $A + C \xrightarrow{K_2} D$ (slow)

Rate will be :-

(1) $K_2[A]\,[C]$

(2) $K_1K_2[A]$

(3) $K_1K_2[A]^2$

(4) $K_1K_2[A][C]$

43) The units of rate constant and rate of reaction are identical for?

(1) Zero order

(2) I^{st} order

(3) 2nd order

(4) Third-order

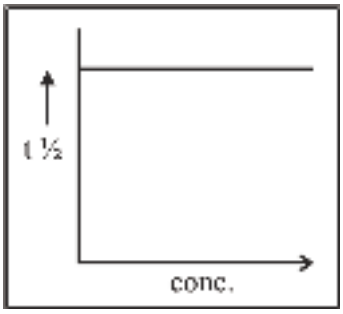
44) If $3A \rightarrow 2B$, then the rate $\left(+\frac{d[B]}{dt}\right)$ is equal to :-

(1) $+2\frac{[A]}{dt}$

(2) $-\frac{1}{3}\frac{d[A]}{dt}$

(3) $-\frac{2}{3}\frac{d[A]}{dt}$

(4) $-\frac{3}{2}\frac{d[A]}{dt}$

45) A graph between $t_{1/2}$ and concentration for n^{th} order reaction is a straight line. Reaction of this nature is completed 50% in 10 minutes when concentration is 2 mol L^{-1}. This is decomposed 50% in t minutes at 4 mol L^{-1}, n and t are respectively 

(1) 0, 20 min.

(2) 1, 10 min.

(3) 1, 20 min.

(4) 0, 5 min.

BIOLOGY

1)

Consider the following four statements (a-d) and select the option which include all the correct ones only regarding Ernst Mayr :
(a) The Darwin of the 20th century.
(b) He is pioneered the currently accepted definition of a biological concept of species
(c) Mayr was awarded nobel prize in the field of Evolutionary biology

(d) His research spanned Ornithology, Taxonomy, systematics, Zoogeography and Evolution.

(1) Statements : (a), (b) and (c)

(2) Statements : (a), (b) and (d)

(3) Statements : (b), (c) and (d)

(4) Statements : (a), (b), (c) and (d)

2) The lowest comman taxonomic categories of Wheat and Mango is

(1) Division

(2) Class

(3) Family

(4) Order

3) Which of the following statement is not correct ?

(1) The word systematic is derived from the Greek word 'Systema' which means identification of unknown organisms.

(2) Classification is the process by which anything is grouped into convenient categories based on some easily observable characters.

(3) Linnaeus's systematics include identification, nomenclature and classification.

(4) *Petunia* and *Datura* are placed in the family Solanaceae.

4) The number of species that are known and described :-

(1) 1.7 - 1.8 billion

(2) 17 - 18 million

(3) 17 - 18 lakh

(4) 1.7 - 1.8 lakh

5) Which option is not related with given diagram : 

(1) They have two nucleus

(2) They have contractile vacuole

(3) They have Anal pore (cytoproct)

(4) They have thousand of flagella

6) Which of the following is the correct order of taxonomic categories of plants, from broadest to narrowest?

(1) Family, Order, Class, Phylum, Kingdom

(2) Kingdom, Phylum, Class, Order, Family

(3) Kingdom, Class, Phylum, Order, Family

(4) Kingdom, Division, Class, Order, Family

7) Binomial system of nomenclature means that every organism has :-

(1) One scientific nomenclature consisting of two specific epithet

(2) A name given by two scientists

(3) One scientific nomenclature consisting of one specific epithet

(4) A number in an international catalogue by which an organism is identified

8)

In which of the following systems of classification give equal weightage to both vegetative and reproductive characters ?
(A) Artificial classification
(B) Natural classification
(C) Numerical classification
(D) Practical classification
Options :-

(1) A and C

(2) B and C

(3) A, B and C

(4) A, B, C and D

9) Match the column :

	Column-I		**Column-II**
a	Phylogenetic system	1	Chemical constituents of plants
b	Numerical taxonomy	2	Based on all observable characters
c	Cytotaxonomy	3	Based on evolutionary relationship
d	Chemotaxonomy	4	Based on chromosome number, structure and behaviour

	a	b	c	d
(1)	3	4	1	2
(2)	3	2	4	1
(3)	2	4	1	3
(4)	3	4	2	1

(1) 1

(2) 2

(3) 3

(4) 4

10) Which of the following is incorrect ?

(1) Genus comprises a group of related species which has more characters in common in comparison to species of other genera

(2) human beings belong to homo species

(3) Potato and brinjal are two different species belonging to the genus *Solanum*

(4) Lion, Tiger, leopard belong to the genus *Panthera*

11) (i) Classification is not a single step process but involves hierarchy of steps in which each step represent a ___A____ or category.

(ii) Category is a part of overall taxonomic arrangement, it is called ___B___
(iii) All category together constitute the ___C_____
(iv) Each category, refered to as a unit of classification represent a rank and is commonly termed as ___D___.
Find out the A, B, C and D are choose the correct.

(1) A - Rank, B - Taxonomic categories,
C - Taxonomic hierarchy, D - Taxon

(2) A - Taxon, B - Taxonomic hierarchy,
C - Rank, D - Taxonomic categories

(3) A - Taxonomic hierarchy, B - Rank,
C - Taxon, D - Taxonomic categories

(4) A - Taxonomic categories, B - Taxon,
C - Taxonomic hierarchy, D - Rank

12) Nomenclature is governed by certain universal rules which one of the following is contrary to the rules of nomenclature ?

(1) Biological names are generally in latin and written in italics

(2) First word in biological name is genus name and second word is specific epithet

(3) Biological names are printed in italics to indicate their italian origin

(4) Genus start with capital letter while the specific epithet start with a small letter

13) *Pisum sativum, Allium cepa, Panthera tigris, Saraca indica, Mangifera indica, Cleom viscosa, Solanum tuberosum, S. melongena and Solanum nigram.*
These are some names of organisms. How many species and genera of plants are included respectively in these names :-

(1) 6, 8

(2) 9, 7

(3) 8, 8

(4) 8, 6

14) *Solanum tuberosum, Solanum melongena* and *Solanum nigrum* represents :-

(1) They are different species of same genus

(2) They are different species of different genus

(3) They are member of same species

(4) *Solanum* is a name of species while *nigrum, melongena* and *tuberosum* represent variety

15) When we move from species to kingdom in taxonomic hierarchy. The common character from one taxon to another taxon :-

(1) Can increase

(2) Can decrease

(3) Always decrease

(4) Can decrease or increase

16) You are given with various categories of classification of housefly. Arrange them in correct sequence of taxonomic hierarchy starting from the lowest category:
(a) Muscidae
(b) insecta
(c) Arthropoda
(d) *Musca*
(e) Diptera
(f) Animalia

(1) f→c→b→e→a→d

(2) d→a→b→c→e→f

(3) d→a→e→b→c→f

(4) d→b→c→a→e→f

17) Which option is related with only monera and not related with other kingdom of R.H. Whittaker?

(1) Decomposer nature

(2) Presence of cell wall and cell membrene

(3) Well develop tissue system

(4) Nitrogen fixation ability with prokaryotic nature

18) What is common between five kingdom and six kingdom system of classification of organisms :-

(1) Number of kingdom for prokaryotes

(2) Both are based on sequence of r-RNA

(3) Number of kingdom for eukaryotes

(4) Number of kingdom for unicellular organisms

19) Read the following statements carefully :-
(A) Some bacteria have resistence to certain antibiotics, this ability is provided by the extra genomic DNA of bacteria
(B) Mesosomes help in respiration and DNA replication
(C) Pili and fimbriae are the surface appendages of bacterial cell which play a role in motility.
(D) Glycocalyx, cell wall and cell membrane form the cell envelope of bacterial cell.
Which of the above statement(s) is/are correct and incorrect ?

	Correct statement	**Incorrect Statement**
(1)	A,B & D	C
(2)	A & B	C & D
(3)	B & D	A & C
(4)	B, C & D	A

(1) 1

(2) 2

(3) 3

(4) 4

20) The bacterial cell is very simple in ___A___ but very complex in ___B___.
Select the option in which A and B are correctly identified.

	A	B
(1)	Behaviour	Structure
(2)	Reproduction	Structure
(3)	Shape	size
(4)	Structure	Behaviour

(1) 1
(2) 2
(3) 3
(4) 4

21) The cell envelop in bacteria consists of:-

(1) Loosely bound three layered structures.
(2) Tightly bound three layered living structures.
(3) Tightly bound three layered non-living and living structures.
(4) Three layered structure with similar function of each layer.

22) Which one of the following option is not related with heterocyst of cyanobacteria ?

(1) Found in BGA like *Nostoc*
(2) Nitrogen fixation ability
(3) CO_2 fixation ability
(4) Thick walled

23) Which set of disease is caused by bacteria?

(1) Cholera, AIDS, tetanus
(2) Cholera, Typhoid, tetanus
(3) Typhoid, tetanus, influenza
(4) AIDS, Mumps, Kuru

24) Match column-I with column-II and select the correct option from the given codes :-

Column-I		Column-II	
(A)	Edible delicious	(i)	*Penicillium*
(B)	Drosophilla of plant kingdom	(ii)	*Ustilago*
(C)	Source of antibioties	(iii)	*Neurospora*

(D)	Smut disease	(iv)	*Mushroom*

(1) A-iv, B-iii, C-ii, D-i

(2) A-iv, B-iii, C-i, D-ii

(3) A-ii, B-iv, C-i, D-iii

(4) A-ii, B-iv, C-iii, D-i

25) Mostly which of the following bacteria are helpful in making curd from milk and production of antibiotics?

(1) Chemoautotophic

(2) Photoautotrophic

(3) Heterotrophic

(4) Symbiotic

26) Consider the following statements :-
(a) They grow in soil, on logs and tree stumps and in living plant bodies as parasites. (b) The mycelium is branched and septate and the asexual spores are generally not found but vegetative reproduction by fragmentation is common. (c) The sex organs are absent but plasmogamy is brought about by fusion of two vegetative or somatic cells of different strains or genotypes. Above statements are related to the member of which class :-

(1) Phycomycetes

(2) Ascomycetes

(3) Basidiomycetes

(4) Deuteromycetes

27) Consider the following statements :-
(a) Presence of phycobilins
(b) No motile stage
(c) Prokaryotic cell structure
(d) Presence of heterocyst
(e) Formation of hormogonia and akinetes
How many of the above statements are only true for Cyanobacteria and are not true for red algae ?

(1) Three

(2) Four

(3) Five

(4) Two

28) What is common among *Porphyra*, *Agaricus*, *Chlorella* and *Spirulina* ?

(1) They all are placed under same kingdom according to five kingdom classification.

(2) They all have nuclear membrane

(3) They all may be used as food or food supplement.

(4) They all are autotrophic

29) **Assertion :** Spores of slime mould can survive for many years even under adverse conditions.

Reason : Slime mould spore posses true cell wall.

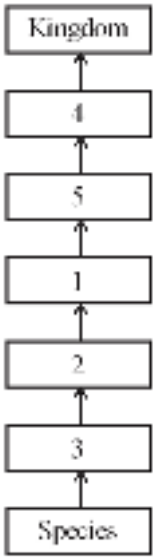
(1) Both Assertion & Reason are True but Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True & the Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

30) In the following flow diagram, identify the correct categories according to taxonomic hierarchy.



(a) Primata, diptera and carnivora belongs to category '1'
(b) Petunia, Datura and Solanum belong to same family and it is category '2'
(c) Angiospermae belongs to category '5'
(d) Man and dog show maximum similarity at category '4'
(e) Category '3' is same for lion, tiger and leopard
Select the correct statements.

(1) a, b, d, e

(2) b, c, d, e

(3) a, b, c

(4) a, b, e

31) Which one is incorrectly matched pair?

(1) Phycomycetes – *Mucor, Albugo*

(2) Ascomycetes – *Penicillium, Aspergillus*

(3) Basidiomycetes – *Puccinia, Agaricus*

(4) Deuteromycetes – *Ustilago, Colletotrichum*

32) Which statement is/are wrong with respect to Archaebacteria.
(a) Archaebacteria differ from other bacteria in having a different cell wall structure.
(b) Feature of cell wall is responsible for their survival in extreme conditions.
(c) Methanogens are present in the guts of several ruminant animals such as cow & buffalo.
(d) These are oldest of the living fossils.
(e) Peptidoglycan & muramic acid are present in cell wall

(1) a and e

(2) b and e

(3) a, b and c

(4) only e

33) Select the correct combination of statements (A–D) regarding the characteristics of certain organisms.
(A) *Mycoplasma* lacks cell wall and can not survive without oxygen.
(B) Ascospores are 8 or 4 in number and endogenously formed.
(C) Fimbriae are larger than flagella
(D) *Nostoc* and *Anabena* are nitrogen fixing autotrophic bacteria found in coralloid root of *cycas*.
The correct statements are :-

(1) A & D

(2) C & D

(3) B & C

(4) B & D

34) Read the following terms carefully.
Cilia, Flagella, Zygote, Water bloom, Photosynthesis, Pellicle, Peptidoglycan wall, Bioluminescence, Multicellular, heterocyst, decomposer, Pseudopodia. How many terms are not related with kingdom Protista:-

(1) 4

(2) 5

(3) 3

(4) 6

35) Select the **incorrect** statement regarding photosynthetic protista :-

(1) Diatoms are the chief producers in the oceans

(2) The spores are dispersed by air currents

(3) The pigments of euglenoids are identical to those present in higher plants

(4) Red dinoflagellate is *Gonyaulax* which causes red tides

36) Read the following two statements (A and B) regarding members of monera and protista.
(A) Their cell wall composition is same
(B) Their cell type is different

(1) Statement A and B are correct

(2) Statement A is correct and B is incorrect

(3) Statement B is correct and A is incorrect

(4) Statement A and B are incorrect

37) Match the column I with column II :-

Particles/Micro organisms	**Diseases**

(i)	Viroid	(a)	Creutzfeldt Jackob diseases (CJD)
(ii)	Virus	(b)	Citrus canker
(iii)	Bacteria	(c)	Potato spindle tuber disease
(iv)	Prion	(d)	Mumps

Options are :-

(1) i - a, ii - b, iii - d, iv - c

(2) i - c, ii - d, iii - a, iv - b

(3) i - c, ii - d, iii - b, iv - a

(4) i - a, ii - c, iii - b, iv - d

38) Match the column-I with column-II and select the correct option :

	Column-I (Organisms)		**Column-II (Features)**
(A)	Dinoflagellates	(i)	Used extensively for genetic work
(B)	Euglenoid	(ii)	Complex structures of cell wall
(C)	Neurospora	(iii)	Stiff cellulose plates
(D)	Thermo-acidophiles	(iv)	Protein rich elastic Pellicle layer

(1) A-(i), B-(ii), C-(iii), D-(iv)

(2) A-(i), B-(iv), C-(iii), D-(ii)

(3) A-(iii), B-(ii), C-(i), D-(iv)

(4) A-(iii), B-(iv), C-(i), D-(ii)



39) Find correct option regarding this diagram :

(a) Colour of this organism appear as yellow, brown, blue and red but not green.

(b) Releases toxins that kills fishes and other marine animal

(c) Two flagella differ in their position

(d) Cell wall is made of peptidoglycan.

(1) a and b

(2) b and c

(3) a, b and c

(4) b, c and d

40) Match Column-I with Column-II and select the correct option from the codes given below :-

Column-I		Column-II	
A	Indestructible wall	i	Euglenoids
B	Red tides	ii	Diatoms
C	Mixotrophic nutrition	iii	Slime moulds
D	Plasmodium	iv	Dinoflagellates

(1) A-(ii),B-(iv), C-(i), D-(iii)

(2) A-(ii),B-(iv), C-(iii), D-(i)

(3) A-(ii),B-(iii), C-(i), D-(iv)

(4) A-(i),B-(iv), C-(iii), D-(ii)

41) In Ascomycetes and Basidiomycetes the intervening period between plasmogamy and karyogamy is known as:

(1) Diplophase

(2) Dikaryophase

(3) Interphase

(4) Resting phase

42)

Arrange the following events in **correct** sequence for sexual cycle of fungi?
(a) Karyogamy
(b) Meiosis
(c) Zygote formation
(d) Plasmogamy
(e) Nucleus pair formation

(1) d, e, a, c, b

(2) d, a, e, c, b

(3) e, a, d, b, c

(4) a, d, e, c, b

43)

List some of the fungi is given below :
Mucor, Rhizopus, Albugo, Aspergillus, Penicillium, Puccinia, Ustilago

In above mention fungi how many of them shows plasmogamy is followed by karyogamy immediately :

(1) Two

(2) Three

(3) Four

(4) Five

44) Choose the incorrect statement about phycomycetes.

(1) Members are found in aquatic habitat also
(2) Spores are endogeneous produced in sporangium
(3) A zygospore is formed by reduction division
(4) They show all type of sexual reproduction

45) *Morels* and *Truffles* are edible fungi belong to which class?

(1) Basidiomycetes
(2) Ascomycetes
(3) Deutromycetes
(4) Phycomycetes

46) **Assertion:** Deuteromycetes are commonly known as imperfect fungi.
Reason: Only asexual or vegetative stages of these fungi are known.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

47) Choose the incorrect statement from following:-

(1) Yeast are used to make bread and bear.
(2) Fungi prefer to grow in warm and humid places.
(3) Cell wall of fungi is composed of chitin and polysaccharides.
(4) Fungi show symbiotic association with algae as mycorrhiza and with roots of higher plants as lichens.

48) Choose the **incorrect** pair of following

(1) Phycomycetes - Obligate parasite
(2) Ascomycetes-Rust and smut disease
(3) Basidiomycetes - Basidiospore produce exogenously
(4) Deuteromycetes - Help in mineral cycling

49) Which of the following is **true** :

(1) Mad cow disease in cattle is due to viroid
(2) Lichen are not found in polluted area
(3) Viroids are infectious agents which do not have nucleic acids
(4) Virus, viroid and prions are classified in kingdom monera by whittaker

50) Choose the incorrect statement for lichen:-

(1) They are very sensitive to SO_2

(2) They are pioneer community in succession on rocks

(3) Algal component called mycobiont and fungal component called phycobiont

(4) Major part of lichen's thallus is composed of fungal component.

51) Few stages of life cycle of moss is given below. Arrange these events in correct order starting from sex organ bearing structure and select the correct option from options given below
A: Protonema
B: Leafy gametophyte
C: Spore formation
D: Spore germination
E: Fertilization
F: Sporophyte

(1) B → E → F → D → C → A

(2) A → B → C → D → E → F

(3) A → B → E → F → C → D

(4) B → E → F → C → D → A

52) The main plant body is a sporophyte and differentiated in true root, stem and leaves, is observed in:

(1) Bryophyta and Pteridophyta

(2) Pteridophyta and Gymnosperm

(3) Algae and Bryophyta

(4) Bryophyta and Gymnosperm

53) Antheridium and archegonium both are present in:

(1) Pteridophyta

(2) Gymnosperm

(3) Angiosperms

(4) Algae

54) Prothallus is
a. Multicellular
b. Sporophyte
c. Usually autotrophic
d. Usually dioecious

(1) a and c

(2) a and b

(3) a and d

(4) b and c

55) Fusion of gametes occurs inside oogonium in :

(1) *Porphyra*

(2) *Funaria*

(3) *Marchantia*

(4) All of the above

56) Fusion of male gamete with female gamete occurs inside archegonium and seeds are produced after fertilization in:

(1) *Pinus* and *Salvinia*

(2) *Cycas* and *Salvinia*

(3) *Pinus* and *Cycas*

(4) *Salvinia* and *Adiantum*

57) Biflagellated male gametes are observed in:

(1) *Pinus*

(2) *Porphyra*

(3) *Marchantia*

(4) *Spirogyra*

58) **Assertion:** In Bryophyta sex organs are present on photosynthetic, green, leafy, multicellular sporophyte
Reason: Zygote in Bryophyta does not undergo reductional division just after its formation

(1) Assertion and reason both are true and reason is correct explanation of assertion

(2) Assertion and reason both are true and reason is not correct explanation of assertion

(3) Assertion is incorrect but reason is correct

(4) Assertion and reason both are incorrect

59) Which of the following correctly explain the statement given below ?
Statement: In Bryophyta and Pteridophyta sporophyte is multicellular

(1) Water is required for fertilization in Bryophyta and Pteridophyta

(2) Gametophyte is multicellular in Bryophyta and Pteridophyta and produced by mitosis in spore

(3) Zygote undergo equational division after its formation in Bryophyta and Pteridophyta

(4) Sporophyte in Bryophyta and Pteridophyta is photosynthetic and autotrophic

60) Isogamy with non-flagellated gametes occurs in:

(1) *Ulothrix*

(2) *Spirogyra*

(3) *Chara*

(4) 1 and 2 both

61) Which of the following is/are heterosporous pteridophytes?
A. Selaginella

B. Pinus
C. Salvia
D. Cycas

(1) A and C
(2) A, B, C and D
(3) Only A
(4) B and D

62) Which of the following is not true regarding Gymnosperms ?

(1) They are heterosporous
(2) They are seeded plants in which ovule is covered by integuments
(3) Archegonia are present inside male cone
(4) 2 and 3 both

63) **Assertion:** *Cycas* is a dioecious plant
Reason: In *Cycas* male cone having microsporophyll and female cone with megasporophylls are present on different trees

(1) A is true but R is false
(2) A and R both are true and R explain A
(3) A and R both are true and R does not explain A
(4) A is false but R is true

64) **Statement A:** The sporophytes bear male and female sex organs called antheridia and archegonia respectively in most of pteridophytes
Statement B: Water is required for transfer of male gametes to archegonium where zygote is formed after fertilisation which divides and produces a sporophyte in Pteridophyta.

(1) Statement and B are true
(2) Statement A and B both are false
(3) Statement A is true but B is false
(4) Statement B is true but A is false

65) In which of the following male and female gametes produce on different plant body?

(1) *Marchantia*
(2) *Pinus*
(3) *Cycas*
(4) 1 and 3 both

66) Agar agar is obtained from:

(1) *Fucus* a brown alga
(2) *Gracillaria* a brown alga
(3) *Gracillaria* a red alga

(4) *Fucus* a red alga

67) Chlorophyll-bearing simple, thalloid, autotrophic and largely aquatic organisms are classified in:

(1) Algae
(2) Pteridophyta
(3) Bryophyta
(4) Gymnosperm

68) Which of the following character does not require for survival of conifers in extremes of environmental conditions ?

(1) presence of motile male gametes
(2) Thick cuticle
(3) Sunken stomata
(4) Needle like leaves

69) Match the column I with column II and select the correct option from options given below:

	Column I		**Column II**
A	Moss	P	*Marchantia*
B	Red algae	Q	*Polysiphonia*
C	Sphenopsida	R	*Polytrichum*
D	Liverwort	S	*Equisetum*

(1) A: R; B: Q; C: S; D: P
(2) A: R; B: Q; C: P; D: S
(3) A: R; B: S; C: Q; D: P
(4) A: P; B: Q; C: S; D: R

70) Which of the following event is precursor to the seed habit in plants?

(1) Formation of only one type of spores
(2) Presence of motile male gametes
(3) Development of zygote into young embryo within the female gametophytes
(4) Presence of vascular bundles

71) Identify the I, ii, iii, and iv in given table and select the correct option from options given below:

Male gametophyte with few cells	i
ii	Unicellular Green alga
iii	Archegonia
Antherozoid in Bryophyta	iv

(1) i: Pollen grain; ii:*Ulothrix*; iii: female reproductive organ; iv: Motile and flagellated

(2) i: Pollen grain; ii:*Chlamydomonas*; iii: female reproductive organ; iv: Motile and flagellated

(3) i: Pollen grain; ii:*Chlamydomonas*; iii: Male reproductive organ; iv: Motile and flagellated

(4) i: Embryo sac; ii:*Chlamydomonas*; iii: female reproductive organ; iv: Motile and flagellated

72) Identify the two correct statements from the given statements and select the correct option from options given below:
a. Algin and carrageen are hydrocolloid obtained from brown algae and red algae respectively
b. Water is required for transfer of antherozoids from antheridium to the mouth of archegonium in gymnosperms
c. The organs in pteridophytes possesses well differentiated vascular tissue
d. Green algae are grass green in colour due to presence of chlorophyll a and chlorophyll c

(1) a and c

(2) a and b

(3) a and d

(4) b and c

73) Study the given diagram and select the correct statement for given diagram:



(1) It is having two types of shoots, long and dwarf shoot

(2) Pollination in this plant occurs by water or by insect

(3) Seeds are surrounded by ovary wall

(4) Male and female gametophyte are independent to sporophyte

74) Identify the correct statements from the following statements and select the correct option from options given below:
a. The male and female cones are borne on different plants in *Pinus*
b. Female gametophytes in gymnosperms bear two or more archegonia
c. The ovules are borne on megasporophylls which may be clustered to form male cone in gymnosperms
d. Branched stem is present in *Pinus* and *Cedrus*

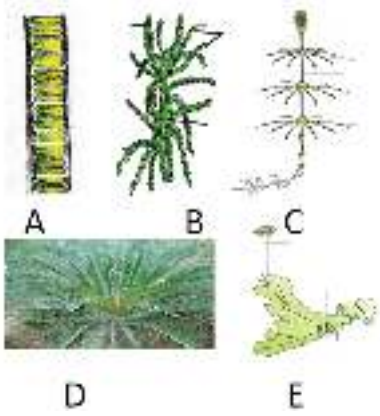
(1) a and c

(2) a and b

(3) b and d

(4) b and c

75) Few diagrams are given below. Arrange these in ascending order of complexity in sporophyte



and select the correct option

(1) A,B,C,D,E
(2) A,E,B,C,D
(3) A,B,E,C,D
(4) D,C,B,E,A

76) **Statement A:** Asexual reproduction in green algae usually by spore, the most common being the zoospores
Statement B: Zoospore in green algae are flagellated and usually have 2 to 8 apical flagella

(1) Statement A and B both are true
(2) Statement A is true but R is false
(3) Statement B is true but A is false
(4) Statement A and B both are false

77) Which of the following is not matched correctly?

(1) Psilopsida: *Psilotum*
(2) Lycopsida: *Lycopodium*
(3) Pteropsida: *Selaginella*
(4) Pteropsida: Adiantum



78) Which of the following is true regarding the given figure ?

(1) It shows oogamy with flagellated male gametes
(2) It produces zoospore during asexual reproduction
(3) Due to pre dominance of r- phycoerythrin its thallus is red in colour
(4) It is a blue green alga which is red in colour

79) Archegonium and antheridium in Bryophyta are:

(1) Multicellular and jacketless

(2) Unicellular and jacketed
(3) Multicellular and Jacketed
(4) Unicellular and jacketless

80) The leaves in pteridophytes are microphylls as in …a…or macrophylls as in ….b..

(1) A: *Selaginella*; B *Lycopodium*
(2) A: *Selaginella*; B *Salvia*
(3) A: *Selaginella*; B Ferns
(4) A: Ferns; B *Selaginella*

81) following fertilization, the zygote does not produce embryo in:

(1) *Fucus*
(2) *Cedrus*
(3) *Pteris*
(4) *Funaria*

82) Pollen grains in gymnosperms are carried away from microsporangia to mouth of ovule by:

(1) Water
(2) Air
(3) Honey Bee
(4) All of the above

83) Which of the following is not associated with naked seed plant?

(1) Homospory
(2) Heterospory
(3) Seed formation
(4) Archegonia

84) Which of the following is not true regarding gemma?

(1) Multicellular
(2) Green photosynthetic
(3) Observed in *Marchantia*
(4) Sexual bud

85) In how many of the following plants water is required for fertilization?
Pinus, Cycas, Funaria, Pteris, Salvia, Cedrus, Ulothrix, Gingko,

(1) Four
(2) Three
(3) Five
(4) Six

86) Two lateral flagella in zoospores are present in:

(1) *Fucus* and *Porphyra*
(2) *Laminaria* and *Ulothrix*
(3) *Fucus* and *Dictyota*
(4) *Porphyra* and *Polysiphonia*

87) Which of the following is used as packing material for trans-shipment of living specimens?

(1) *Sphagnum*
(2) *Funaria*
(3) *Polytrichum*
(4) *Marchantia*

88) Evolutionary these are the first plants to possess vascular plants. They are:

(1) Bryophytes
(2) Pteridophytes
(3) Gymnosperms
(4) Algae

89) One of the tallest plant is:

(1) *Cedrus*
(2) *Pinus*
(3) *Sequoia*
(4) *Cycas*

90) Pinnately compound leaves and coralloid roots are present in:

(1) *Cycas*
(2) *Pteris*
(3) *Selaginella*
(4) *Pinus*

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	2	3	2	4	1	1	3	1	3	1	1	3	2	3	1	2	4	4	3
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	2	4	1	3	2	2	3	4	4	1	1	2	4	3	1	1	4	1	2
Q.	41	42	43	44	45															
A.	2	4	3	1	1															

CHEMISTRY

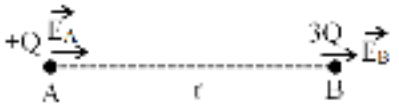
Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	4	2	3	3	3	2	2	1	3	1	4	2	1	1	4	1	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	2	3	1	2	4	2	4	4	4	1	4	1	2	3	3	3	3	3	3
Q.	86	87	88	89	90															
A.	3	3	1	3	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	2	1	1	3	4	4	3	1	2	2	1	3	4	1	3	3	4	3	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	2	2	3	3	1	3	1	4	4	4	4	3	2	3	3	4	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	2	3	2	1	4	2	2	3	4	2	1	1	1	3	3	3	3	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	3	1	4	4	3	1	1	1	3	2	1	1	3	2	1	3	3	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	2	1	4	2	3	1	2	3	1										

SOLUTIONS

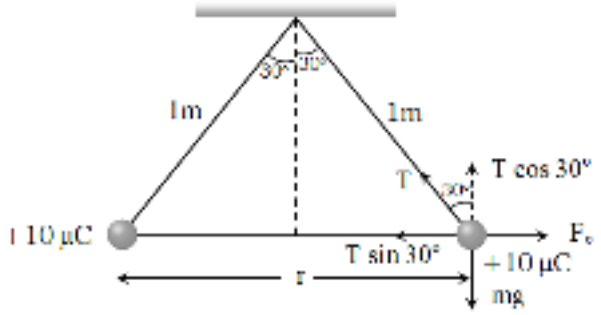
PHYSICS



1)

$$\vec{E}_A = \frac{k(3Q)}{r^2}\left(+\hat{i}\right) = \vec{E}\,\vec{E}_B = \frac{k(Q)}{r^2}\left(+\hat{i}\right) = \vec{E}/3$$

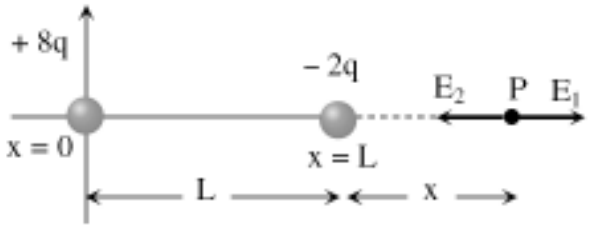
2) In the following figure, in equilibrium $F_e = T \sin 30^\circ$, r = 1m



$$\Rightarrow 9\times10^9.\frac{Q^2}{r^2} = T\times\frac{1}{2}$$

$$\Rightarrow 9\times10^9.\frac{(10\times10^{-6})^2}{1^2} = T\times\frac{1}{2} \Rightarrow T = 1.8\ N$$

3) The net field will be zero at a point outside the charges and near the charge which is smaller in magnitude.



Suppose E.F. is zero at P as shown.

$$\frac{x}{L+x} = \sqrt{\frac{2}{8}} \Rightarrow \frac{x}{L+x} = \frac{1}{2}$$

2x = L + x

x = L

So distance of P from origin is L + L = 2L.

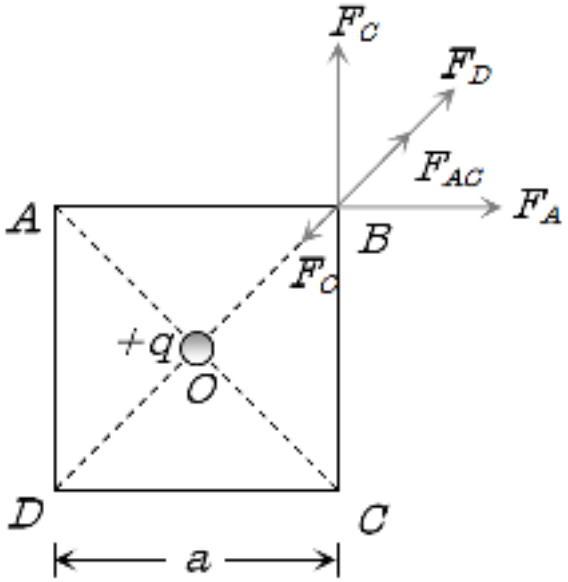
4) If all charges are in equilibrium , system is also in equilibrium.

Charge at centre : charge q is in equilibrium because no net force acting on it

Corner charge : If we consider the charge at corner B.

This charge will experience following forces

$$F_A = k\frac{Q^2}{a^2}, F_C = \frac{kQ^2}{a^2},$$



$F_D = \frac{kQ^2}{(a\sqrt{2})^2}$ and $F_O = \frac{KQq}{(a\sqrt{2})^2}$ Force at B away from the centre $= F_{AC} + F_D$

$= \sqrt{F_A^2 + F_C^2} + F_D = \sqrt{2}\frac{kQ^2}{a^2} + \frac{kQ^2}{2a^2} = \frac{kQ^2}{a^2}\left(\sqrt{2} + \frac{1}{2}\right)$

Force at B towards the centre $= F_O = \frac{2kQq}{a^2}$

For equilibrium of charge at B, $F_{AC} + F_D = F_O$

$\Rightarrow \frac{KQ^2}{a^2}\left(\sqrt{2} + \frac{1}{2}\right) = \frac{2KQq}{a^2}$

$\Rightarrow q = \frac{Q}{4}\left(1 + 2\sqrt{2}\right)$

5) Total charge $Q = 80 + 40 = 120\,\mu C$. By using the formula $Q_1' = Q\left[\frac{r_1}{r_1 + r_2}\right]$. New charge on sphere A is $Q_A' = Q\left[\frac{r_A}{r_A + r_B}\right] = 120\left[\frac{4}{4+6}\right] = 48\,\mu C$.

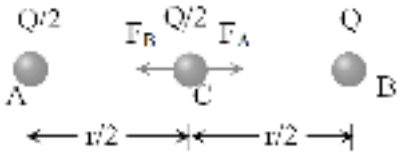
Initially it was $80\mu C$ i.e., $32\,\mu C$ charge flows from A to B.

6) Initially



$F = k\frac{Q^2}{r^2}$ .....(i)

Finally



Force on *C* due to *A*, $F_A = \frac{k(Q/2)^2}{(r/2)^2} = \frac{kQ^2}{r^2}$

Force on *C* due to *B*, $F_B = \frac{KQ(Q/2)}{(r/2)^2} = \frac{2KQ^2}{r^2}$

 Net force on C, $F_{net} = F_B - F_A = \frac{kQ^2}{r^2} = F$

7)

Due to transfer to electrons; mass also changes.

8) $Q = \pm ne$ and charge lesser than 1 C is possible.

9) (A) $\frac{dy}{dx} = \cos x + \sin x$

at $x = \frac{\pi}{2}$, $\frac{dy}{dx} = \cos\frac{\pi}{2} + \sin\frac{\pi}{2} = 1$

(B) $\frac{dy}{dx} = 7e^{7x}$

$x = 0$, $\frac{dy}{dx} = 7$

(C) $\frac{dy}{dx} = \frac{1}{x}$

at $x = 1$, $\frac{dy}{dx} = 1$

(D) $\frac{dy}{dx} = 1$

10) $\int_0^2 5\,dx = 5[x]_0^2 = 5[2 - -0] = 10$

$\int_1^2 x dx = \left[\frac{x^2}{2}\right]_1^2 = \left[\frac{4}{2} - \frac{1}{2}\right] = 1.5$

$\int_1^2 x^{-1}\,dx = \left[\log_e x\right]_1^2 = \log_e 2 - \log_e 1 = \log_e 2$

$\int_0^\infty e^{-x}\,dx = \left[-e^{-x}\right]_0^\infty = \left[e^o - e^{-\infty}\right] = 1 - 0 = 1$

11)

$y = 3x + 4$
$y = mx + c$
$m = 3$ (+ve slope)
$c = 4$ (+ve y intercept)
Hence option (1) is correct.

12)

$\tan(210°) = \tan(\pi + 30°)$
$= \tan 30°$
$= \frac{1}{\sqrt{3}}$

13) **QUESTION EXPLANATION :**

Two identical particles each having mass m and charge q are placed at a distance r. The electrostatic force of repulsion between them is equal to the gravitational force of attraction. The specific charge (q/m) is to be determined.

CONCEPT :
Comparison of Coulomb's force and gravitational force

FORMULA : Electrostatic force: $F_e = \left(\frac{1}{4\pi\varepsilon_0}\right) \times \left(\frac{q^2}{r^2}\right)$

Gravitational force: $F_g = G \times \left(\frac{m^2}{r^2}\right)$

SOLUTION :
Given condition:
$F_e = F_g$

$$\left(\frac{1}{4\pi\varepsilon_0}\right) \times \left(\frac{q^2}{r^2}\right) = G \times \left(\frac{m^2}{r^2}\right)$$

Cancel r^2 from both sides: $\left(\frac{1}{4\pi\varepsilon_0}\right) \times q^2 = G \times m^2$

Rearranging:
$$\left(\frac{q^2}{m^2}\right) = 4\pi\varepsilon_0 \times G$$
$$\Rightarrow \frac{q}{m} = \sqrt{(4\pi\varepsilon_0 G)}$$

FINAL ANSWER : Option (3)

14) **Explain question:**
Charges at A: +1 μC, at B: – 1 μC, at C: + 2μC on an equilateral triangle of side 0.1 m. Find the resultant force on charge at C.

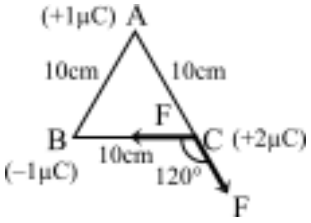
Concept:
Coulomb's law and vector addition in an equilateral triangle.

Solution:

$$F_A = F_B = F = \frac{k(1)(2) \times 10^{-12}}{\left(10^{-1}\right)^2}$$

$F = 9 \times 10^9 \times 2 \times 10^{-10}$

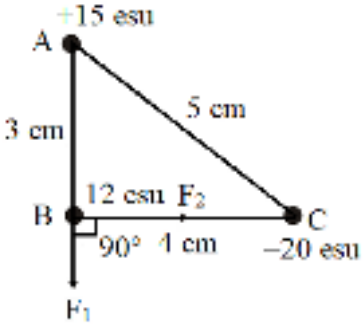
$F = 1.8N$



$(F_{net})_c = F = 1.8N$

Final Answer: Option (2)

15) **Explain Question:**

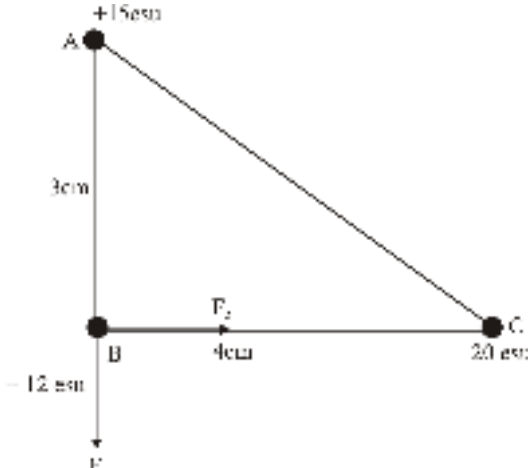


We have to calculate net force acting on 'B'.

Concept:

Electrostatic force.

Solution:



$$F_1 = \frac{k(12)(15)}{3^2} = \frac{1 \times 12 \times 15}{9}$$

$F_1 = 20$ dyne

$$F_2 = \frac{k(12)(20)}{4^2} = \frac{1 \times 12 \times 20}{16}$$

$F_2 = 15$ dyne

$$(F_{net})_B = \sqrt{F_1^2 + F_2^2} = \sqrt{(20)^2 + (15)^2}$$

= 25 dyne

Final Answer: Option (3)

16) **Explain the Question :** Two spheres of same mass and same length carry charges +q + and 4q. They repel and reach equilibrium. Which diagram is correct ?
Concept : This question is based on equilibrium of forces on each charged sphere.
Solution : Each sphere experiences
weight mg downward
tension along the string
electrostatic force F_e horizontally
Electrostatic force magnitude on both spheres is the same :

$$F_e = \frac{k(4q)\ (q)}{r^2}$$

For the +q sphere :

$$\tan\theta_1 = \frac{F_e}{mg}$$

For the +4q sphere :

$$\tan\theta_2 = \frac{F_e}{mg}$$

Since both have the same mass and the same horizontal force,
$\theta_1 = \theta_2$
So both spheres must deflect by equal angles from the vertical.
Final Answer: (1)

17)

Question Explanation : Two bodies are rubbed with each other, one gets positive charge and the other gets negative charge. we need to identify the body having higher work function.

Concept : Methods of Charging of two bodies **(**Work function & electron transfer)

Solution

Work function is the **minimum amount of energy required to remove an electron from the surface of a material**.
On rubbing, **electrons transfer** from one body to the other.
The body that **loses electrons** becomes **positively charged** → it has **lower work function** (electrons escape easily).
The body that **gains electrons** becomes **negatively charged** → it has **higher work function** (holds electrons tightly).

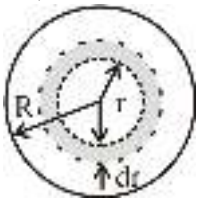
Given:
A becomes **positive** ⇒ A lost electrons ⇒ **lower work function**
B becomes **negative** ⇒ B gained electrons ⇒ **higher work function**

Final Answer : option (2)

18) **Explain the question :** A charged water drop of radius r with charge equal to one electron is in equilibrium under gravity and electric field. Find field intensity.
Concept : This question is based on electrostatics equilibrium.,
Solution :

• Weight of drop : W = mass·g = $\left(\frac{4}{3}\pi r^3\rho\right)g$
• Electric force : $F_e = qE = eE$
• Equilibrium condition : $F_e = W \Rightarrow eE = \frac{4}{3}\pi r^3\rho g$
•Therefore, $E = \frac{4\pi r^3\rho}{3e}g$
Final answer : 4

19) Let us assume an elemental ring of radius 'r' & thickness 'dr', then charge on ring



dQ = σ (Area of shaded region)
= σ (2πrdr)

$$\int dQ = \int \sigma_0 \frac{r^2}{R^2} 2\pi r\, dr \Rightarrow Q = \frac{\pi\sigma_0 R^2}{2}$$

Hence option (4)

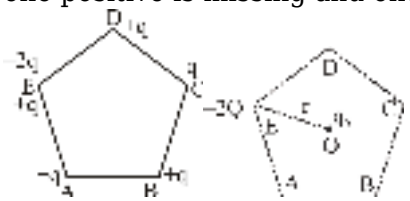
20) $x = \sqrt{3}R$

$$E = \frac{kQx}{(R^2 + x^2)^{3/2}}$$

$$E = \frac{kQ}{(R^2 + 3R^2)^{3/2}}\sqrt{3}R$$

$$E = \frac{kQ}{8R^3}\sqrt{3}R = \frac{\sqrt{3}}{8}\frac{kQ}{R^2}$$

21) Since, it is a regular polygon If all charges were positive net force would be zero. Now one positive is missing and one additional –Q is placed at point E.



22) $E_0 = \sqrt{E_1^2 + E_2^2}$

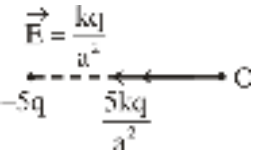
$$E_0 = \sqrt{\left(\frac{\sqrt{2}\,k(8\lambda)}{R}\right)^2 + \left(\frac{\sqrt{2}\,k(6\lambda)}{R}\right)^2}$$

$$E_0 = \frac{\sqrt{2}\,k\,\lambda}{R}\sqrt{8^2+6^2}$$

$$E_0 = \sqrt{2}\frac{k\,\lambda}{R}(10)$$

$$E_0 = 10\sqrt{2}\frac{k\,\lambda}{R}$$

23)



$$\vec{E}_0 = \frac{6kq}{a^2} = 6\vec{E}$$

24) $\vec{P}$ is parallel to $\vec{Q}$ so their resultant is equal to P + Q

25) Two vectors $\vec{A} = a\hat{i} + b\hat{j}$ and $\vec{B} = x\hat{i} + y\hat{j}$ will be mutually perpendicular if $\vec{A}.\vec{B} = 0$

⇒ ax + by = 0 ⇒ ax = – by

which is possible if x = b and y = –a

or x = – b and y = a

 x = 3sinθ and y = 5cosθ

Thus $\vec{B} = 3\sin\theta\hat{i} + 5\cos\theta\hat{j}$

26) Resultant of three forces can be zero if magnitude of third force must present between maximum & minimum value of resultant of two forces.

27) $F^2 = F^2 + F^2 + 2FF\cos\theta$

⇒ $-F^2 = 2F^2\cos\theta$

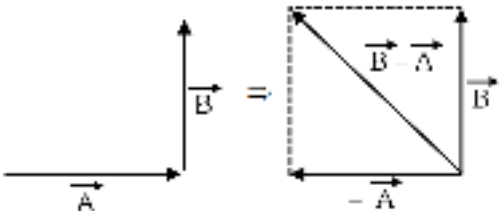
⇒ $\cos\theta = -\frac{1}{2} \Rightarrow \theta = 120^\circ$

28) $\left|\hat{a} + \hat{b}\right| = \sqrt{3}$

$$\sqrt{1^2 + 1^2 + 2.1.1\cos\theta} = \sqrt{3}$$

⇒ 2 cos θ = 1 ⇒ cos θ = $\frac{1}{2}$ ⇒ θ = 60°

$$(2\hat{a} - 5\hat{b}).(3\hat{a} + \hat{b})$$
$$= 6(\hat{a}.\hat{a}) + 2(\hat{a}.\hat{b}) - 15(\hat{b}.\hat{a}) - 5(\hat{b}\cdot\hat{b})$$
$$= 6 + 2|\hat{a}|\,|\hat{b}|\cos 60^\circ - 15|\hat{b}||\hat{a}|\cos 60^\circ - 5$$
$$= 6 + 1 - \frac{15}{2} - 5 = -\frac{11}{2}$$

29) 

30) Cross product gives vector quantity and scalar product gives scalar quantity so both can't compare.

31) $(4\hat{i}-\hat{j})+(-3\hat{i}+2\hat{j})+(-\hat{k})$

$\hat{i}+\hat{j}-\hat{k}$

Unit vector along $\vec{R},\ \hat{R}=\frac{\vec{R}}{|R|}=\frac{(\hat{i}+\hat{j}-\hat{k})}{\sqrt{3}}$

32) Using slope of the curve we can write equation

$a = -10x$

Comparing it with $a = -\omega^2 x$

$\omega^2 = 10$

$\omega = \sqrt{10}$

$\text{frequency } f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{10} = \frac{\sqrt{2.5}}{\pi}s^{-1}$

33) From spring block system,

Time period is given by - $T = 2\pi\sqrt{\frac{m}{k}}$

Here, $t_1 = 2\pi\sqrt{\frac{m}{k_1}}$ and $t_2 = 2\pi\sqrt{\frac{m}{k_2}}$

Now, for series combination of spring,

$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} \Rightarrow T = 2\pi\sqrt{\frac{m}{k}}$

$T^2 = 4\pi^2\frac{m}{k} = 4\pi^2\left(\frac{m}{k_1}+\frac{m}{k_2}\right) = {t_1}^2 + {t_2}^2$

34) The total energy E of a particle vibrating in SHM is given by

$E = \frac{1}{2}m\omega^2 a^2 \ldots (1)$

The kinetic energy K is given by

$K = \frac{1}{2}m\omega^2\left(a^2 - y^2\right)$

Where y = displacement of the particle, but

$K = \frac{E}{2} = \frac{1}{2}\left[\frac{1}{2}m\omega^2 a^2\right]$

$\Rightarrow \therefore \frac{1}{2}\left[\frac{1}{2}m\omega^2 a^2\right] = \frac{1}{2}m\omega^2\left(a^2 - y^2\right)$

or $\frac{a^2}{2} = a^2 - y$

or $y^2 = \frac{a^2}{2}$

$\Rightarrow \therefore y = \frac{a}{\sqrt{2}}$

$\Rightarrow \therefore y = \frac{4}{\sqrt{2}} = 2\sqrt{2}cm$

35) Initially, $T = 2\pi\sqrt{\frac{M}{k}}$ and

Finally, $\frac{5T}{3} = 2\pi\sqrt{\frac{M+m}{k}}$

$\frac{5}{3} \times 2\pi\sqrt{\frac{M}{k}} = 2\pi\sqrt{\frac{M+m}{k}}$

or $\frac{25}{9}\frac{M}{k} = \frac{M+m}{k}$

or 9m + 9M = 25 M

or $\frac{m}{M} = \frac{16}{9}$

36) Comparing $a + 4\pi^2 x = 0$ with standard equation of simple harmonic motion

$a + \omega^2 x = 0$

$\omega^2 = 4\pi^2, \omega = 2\pi$

$2\pi n = 2\pi, n = 1Hz$

37) In first case, springs are connected in parallel, so their equivalent spring constant

$k_p = k_1 + k_2$

So, frequency of this spring-block system is

$f_p = \frac{1}{2\pi}\sqrt{\frac{k_p}{m}}$

or $f_p = \frac{1}{2\pi}\sqrt{\frac{k_1 + k_2}{m}}$

but $k_1 = k_2 = k$

$\therefore f_p = \frac{1}{2\pi}\sqrt{\frac{2k}{m}}$ ... ..(i)

Now in second case, springs are connected in series, so equivalent spring constant

$k = \frac{k_1 k_2}{k_1 + k_2}$

Hence frequency of this arrangement is given by

$f_s = \frac{1}{2\pi}\sqrt{\frac{k_1 k_2}{(k_1 + k_2)m}}$

or $f_s = \frac{1}{2\pi}\sqrt{\frac{k}{2m}}$ ... (ii) $[K_1 = K_2 = K]$

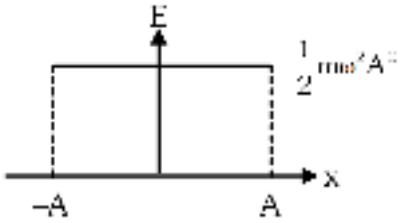
Dividing eq (ii) by (i), we get

$\frac{f_s}{f_p} = \frac{\frac{1}{2\pi}\sqrt{\frac{k}{2m}}}{\frac{1}{2\pi}\sqrt{\frac{2k}{m}}} = \sqrt{\frac{1}{4}}$

or $\frac{f_s}{f_p} = \frac{1}{2}$

38) Given :
displacement $x = 2 \times 10^{-2}\cos\pi t$
Velocity $v = \frac{dx}{dt} = -2 \times 10^{-2}\pi\sin\pi t$
For the first time when
$v = v_{max}, \sin\pi t = 1$
$\sin\pi t = \sin\frac{\pi}{2}$
or $\pi t = \frac{\pi}{2} t = \frac{1}{2}s = 0.5s$

39) Total energy
$E = \frac{1}{2}m\omega^2A^2$
It does not varry with position i.e. x



40) Let $T = 2\pi\sqrt{\frac{l}{g}} \Rightarrow T = 2\pi\sqrt{\frac{0.1}{10}} = \frac{\pi}{5}s$
Time taken by the pendulum in going from A to $B = \frac{T}{4}$
Time taken by the pendulum in going from B to $C = \frac{T}{12}$
Time period of the pendulum, $= 2\left(\frac{T}{4} + \frac{T}{12}\right) = \frac{2T}{3} = \frac{2}{3} \cdot \frac{\pi}{5} = \frac{2\pi}{15}s$

41) Using the general equation of SHM
$x = A \sin \omega t$
as per graph A = 1, T = 8 sec
$\omega = \frac{2\pi}{T} = \frac{\pi}{4} \Rightarrow x = \sin\left[\frac{\pi}{4}t\right]$
acceleration $a = \frac{d^2x}{dt^2} \Rightarrow a = -\frac{\pi^2}{16}\sin\left[\frac{\pi}{4}t\right]$
at $t = \frac{2}{3}$
$a = -\frac{\pi^2}{16}\sin\left(\frac{\pi}{4}.\frac{2}{3}\right) = -\frac{\pi^2}{16}\sin\left(\frac{\pi}{6}\right)$
$a = -\frac{\pi^2}{16} \times \frac{1}{2} = -\frac{\pi^2}{32}\frac{cm}{s^2}$

42) The equation can be written as

$$y = 4\sin\left(4\pi t - \frac{\pi x}{16}\right)$$

Wave velocity is given by

$$v = \frac{\text{coefficient of } t\,(\omega)}{\text{coefficient of } x\,(k)}$$

$$\Rightarrow v = \frac{4\pi}{\frac{\pi}{16}} = 64\frac{cm}{s} \text{in} + ve \;\; x - \text{direction.}$$

43) Given frequency f = 500Hz,
velocity v = 350m/s,
phase difference ($\Delta\phi$) = 60°

By the formula, $\lambda = \frac{v}{f} = \frac{350}{500} = 0.7m$

Also we know that $\Delta x = \Delta\phi\frac{\lambda}{2\pi}$

$$= 60 \times \frac{\pi}{180} \times \frac{0.7}{2\pi} \approx 0.12m \approx 12cm$$

44)

60 π cm s^{-1}

45) Potential energy at extreme $= \frac{1}{2}KA^2 = \frac{1}{2}m\omega^2A^2$

$$= \frac{1}{2}m\left(\frac{g}{\ell}\right)A^2$$

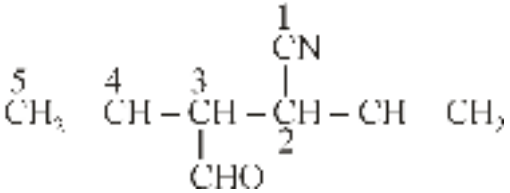
$$= \frac{MgA^2}{2L}$$

CHEMISTRY

46) **Question Explanation :** Identify the IUPAC name of given compound.

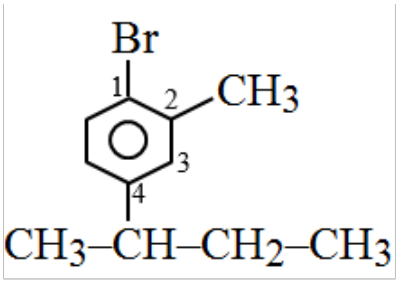
Concept : IUPAC nomenclature of hydrocarbon derivatives.

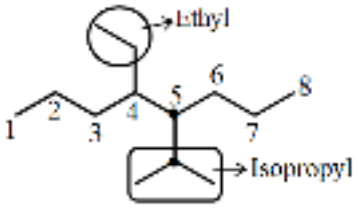
Solution :
Functional group priority ⇒ CN > –CHO



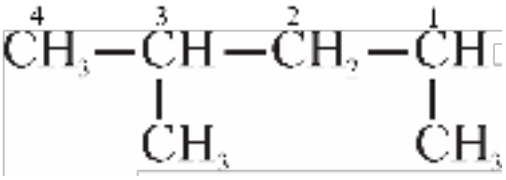
 2-Ethenyl-3-formyl pent-4-ene nitrile.

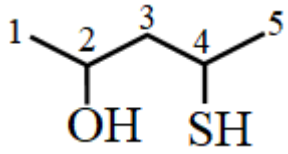
Final Answer : Option (2)

47) 

48) 

4-Ethyl-5-isopropyl octane

49) 

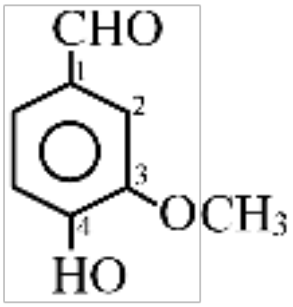
50) 

51)

NCERT XI unit 8

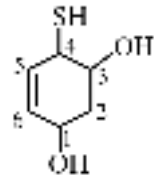
52)

F.G. Should be same

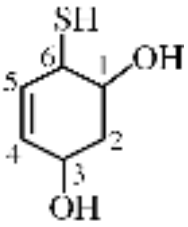
53) 

54)

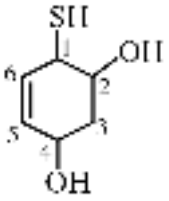
Explain the Question : Identify the structure with correct numbering for the IUPAC name of the given compound.
Concept : This question is based on IUPAC nomenclature, numbering rule.
Solution : Number given by following priority ⇒ PFG > MB > SS

(1) 

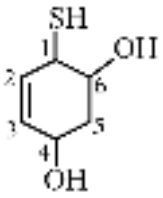
Incorrect numbering
MB did not get Lowest number

(2) 

Correct numbering

(3) 

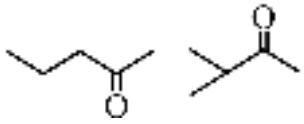
Incorrect numbering PFG did not get Lowest number

(4) 

Incorrect numbering PFG do not get

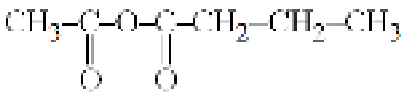
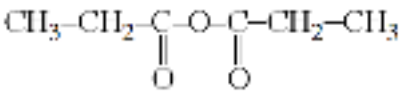
Compound (2) with correct numbering for IUPAC name.

Final answer : Option (2)

55) 

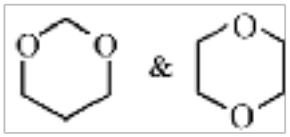
56)

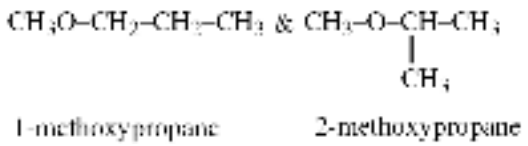
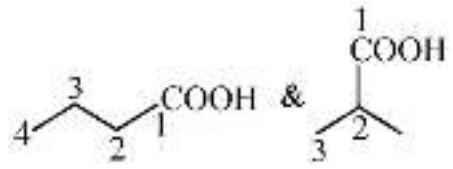
defination based

57)  & 
are Metamers

58)

Explain the Question : Determining which pair of compounds are not positional isomers.
Concept : This question is based on positional isomerism.
Solution :

Given compounds	Remark
1  Butan-1-ol Butan-2-ol	Positional isomers.
2  1,3-dioxane 1,4-dioxane	Positional isomers or metamer.

3 	Positional isomers or metamer.
4 	Both are chain isomers.

Final answer : 4

59) 

60)

M.F is same for both compounds

61)

$CnH_{2n-2}O_3$

62) NCERT-XI, Pg. # 256

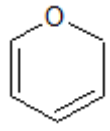
63) NCERT-XI, Pg. # 377

64)

NCERT / XII / Aldehyde & Ketone

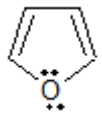
65)

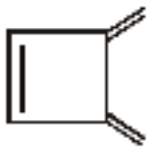
Pyran is 6 member Heterocyclic

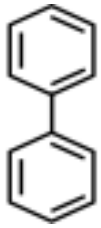


C_5H_6O

Furan is 5 memb heterocyclic and aromatic ring with 4 carbon atom



66)  All C-atom are sp^2

67) 

All carbon are sp^2 hybridised.

68)

Order = $\frac{1}{2}+\frac{1}{2}-1=0$

69)

Rate constant does not depends on conc. of reactant.

70) $xA \rightarrow yB$

$$ROR = \frac{-1}{x}\frac{d[A]}{dt} = \frac{1}{y}\frac{d[B]}{dt}$$

$$\Rightarrow \frac{-1}{x}\frac{d[A]}{dt} = \frac{1}{y}\frac{d[B]}{dt}$$

$$-\frac{d[A]}{dt} = \frac{x}{y}\frac{d[B]}{dt}$$

Taking long on both sides

$$\log\left[-\frac{d[A]}{dt}\right] = \log\left[\frac{d[B]}{dt}\right]\frac{x}{y}$$

$$\log\left[-\frac{d[A]}{dt}\right] = \log\frac{x}{y} + \log\frac{d[B]}{dt} \quad ....(i)$$

from question

$$\log\left[-\frac{d[A]}{dt}\right] = \log\frac{d[B]}{dt} + \log 2 \quad ....(ii)$$

compare the equation (i) & (ii)

$$\frac{x}{y} = 2$$

x : y = 2 : 1

71) $2SO_2(g) + O_2(g) \overset{pt}{\rightarrow} 2SO_3(g)$

$r = k[SO_2]\,[SO_3]^{-1/2}$

overall order of reaction = $1-\frac{1}{2}=\frac{1}{2}$

order with respect to SO_2 = 1

As $[SO_3]$ increases, rate decreases as it has negative order

unit of $k = \left(\frac{mol}{L}\right)^{1-n} sec^{-1} = \left(\frac{mol}{L}\right)^{\frac{1}{2}} sec^{-1} = mol^{1/2}L^{-1/2}sec^{-1}$

72)

From Exp-1 → Exp-2 (B constant = 0.1 M):
[A] : 0.1 → 0.4 (×4)
Rate : $6\times10^{-4} \rightarrow 12\times10^{-4}$ (×2)
⇒ $4^m = 2 \Rightarrow m = 1/2$

From Exp-2 → Exp-3 (A constant = 0.4 M):
[B] : 0.1 → 0.2 (×2)
Rate : $12\times10^{-4} \rightarrow 48\times10^{-4}$ (×4)
⇒ $2^n = 4 \Rightarrow n = 2$

Rate equation: $r = k[A]^{1/2}[B]^2$

73)

Unit of rate = mol L^{-1} s^{-1}
→ For overall order 2,
unit of k = L mol^{-1} s^{-1} or mol^{-1} L s^{-1}, which matches option 2 → correct

74)

$$2A_{(g)} \rightarrow B_{(g)} + 5C_{(g)}$$

Initial	80 mm Hg	0	0

At t = 5 min (80 - 2x)mm Hg x mm Hg 5 x mm Hg
Pt = 80 – 2x + x + 5 x
Pt = 80 + 4x
Pt = 100 mm Hg
4x = 100 - 80
$x = \frac{20}{4} = 5 \text{ mms Hg}$
$(P_A)_{t=0} = 80 \text{ mm Hg}$
$(P_A)_{t=5} = (80 - 2 \times 5) \text{ mm Hg} = 70 \text{ mm Hg}$
rate of disappearance of $A = \frac{-dA}{dt}$
$= \frac{-((PA)_{t=5} - (PA)_{t=0})}{5-0}$
$= \frac{-(70-80) \text{ mm Hg}}{5}$
$= \frac{10}{5} = 2 \text{ mm Hg min}^{-1}$

75)

$r = K_2 \times [N_2O_2] \times [H_2]$

$$keq = \frac{[N_2O_2]}{[NO]^2}$$

$\therefore r = k \times [NO]^2 \times [H_2]$

76) R.O.R = $K[A_2]^a[B_2]^b$

$0.04 = K(0.2)^a(0.2)^b$ ... (i)

$0.04 = K(0.1)^a(0.4)^b$ ... (ii)

$0.08 = K(0.2)^a(0.4)^b$ ... (iii)

$\frac{(i)}{(iii)} \frac{1}{2} = \left(\frac{1}{2}\right)^b$; $b = 1$

$\frac{(i)}{(iii)} \frac{1}{2} = 2^a \times \left(\frac{1}{2}\right)^b$; $a = 1$

77) $\frac{-\Delta[A]}{\Delta t} = -\frac{(4-5)}{5} = \frac{+1}{5} = 0.2$ mol lit^{-1} sec^{-1}

$$\frac{1}{n_1}\left\{\frac{-\Delta(A)}{\Delta t}\right\} = \frac{1}{m_2}\left\{\frac{+\Delta[D]}{\Delta t}\right\}$$

$$\frac{+\Delta[D]}{\Delta t} = \frac{m_2}{n_1}\left\{\frac{-\Delta(A)}{\Delta t}\right\} = \left(\frac{m_2}{n_1} \times 0.2\right) M\,sec^{-1}$$

78) 36 g $O_2 = \frac{36}{32} = 1.125$ mol/min

$$-\frac{1}{2}\frac{[H_2O_2]}{dt} = \frac{d[O_2]}{dt}$$

$\therefore -\frac{d[H_2O_2]}{dt} = 2 \times \frac{d[O_2]}{dt} = 2 \times 1.125$

$= 2.25$ mol/min

79)

Given

reaction $2NH_3 \rightarrow N_2 + 3H_2$

$$\frac{-d[NH_3]}{dt} = K_1[NH_3], \quad \frac{d[N_2]}{dt} = K_2[NH_3], \quad \frac{d[H_2]}{dt} = K_3[NH_3]$$

$$ROR = \frac{1}{2}\frac{-d[NH_3]}{dt} = \frac{d[N_2]}{dt} = \frac{1}{3}\frac{d[H_2]}{dt}$$

$$= \frac{1}{2}K_1[NH_3] = K_2[NH_3] = \frac{K_3(NH_3)}{3}$$

$$\frac{K_1}{2} = K_2 = \frac{K_3}{3}$$

$3K_1 = 6K_2 = 2K_3$

80) $\text{Unit of } K = \frac{\text{Unit of rate}}{\text{Unit of } [A]^x}$

$$= \left(\frac{\text{mol}}{\text{L}}\right)^{1-x} \times \text{time}^{-1}$$

81)

Temp. is not changing so heat released and absorbed is equal.
(1) A + 3B → P :OH = –2x KJ
(2) 2M → 4Q + R :OH = 2x KJ

for (1) : (rate) Rxn = $\frac{1}{3} r_B$ $\frac{1}{3}r_B = \frac{1}{4}r_Q$
for (2) : (rate) Rxn = $\frac{1}{4} r_Q$

$r_B = \frac{3}{4} r_Q$
$r_Q = \frac{4}{3}.y$

82)

$r = K[A]^x [B]^y$
By I, II
A = doubled, B = same, r = doubled
 x = 1
By II, III
A = same, B = doubled, r = doubled
 y = 1
$r = K [A] [B]$
By I, $2 \times 10^{-4} = K \times (1 \times 10^{-2}) \times (2 \times 10^{-2})$
K = 1

83) $0.48 + \log\left\{-\frac{d[A]}{dt}\right\} = \log\left\{+\frac{d[B]}{dt}\right\} + 0.7$
$\log 3 + \log r_A = \log r_B + \log 5$
$\log (3 \times r_A) = \log (5 \times r_B)$
$3r_A = 5r_B$; $\frac{r_A}{5} = \frac{r_B}{3}$ x : y = 5 : 3

84)

Explain Question : Relate dC_A/dt, dn_A/dt and dP_A/dt for A(g) → products at constant V.
Concept : This question is based on different forms of rate
Explanation : $P_AV = n_A \cdot R \cdot T \Rightarrow P_A = (n_A/V) \cdot R \cdot T$.
But $C_A = n_A/V$, so $P_A = C_A \cdot R \cdot T$.
Differentiate: $dC_A/dt = (1/RT) \cdot dP_A/dt$.
Also $C_A = n_A/V \Rightarrow dC_A/dt = (1/V) \cdot dn_A/dt$.
Thus $dC_A/dt = (1/RT) \cdot dP_A/dt = (1/V) \cdot dn_A/dt$.

85)

According to the unit of K reaction is 3rd order
 $r = K[A]^3$

86)

$$\text{Average rate} = \frac{\Delta V}{\Delta t}$$
$$= \frac{V_3 - 0}{40 - 0} = \frac{V_3}{40}$$

87)

$$k_1 = \frac{[C]}{[A]} \Rightarrow [C] = k_1[A]$$
Rate = $k_2[A]\, k_1\, [A] = k_1 k_2\, [A]^2$

Final answer : Rate = $K_1K_2[A]^2$

88)

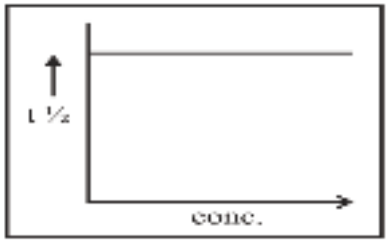
For zero order
$r = k[A]^0$
$$r = k = \frac{\text{mol}}{L \times s}$$
unit of r = unit of k = mol L^{-1} s^{-1}

89) $3A \rightarrow 3B$
$$-\frac{1}{3}\frac{d[A]}{dt} = \frac{1}{2}\frac{d[B]}{dt}$$
$$\frac{d[B]}{dt} = -\frac{2}{3}\frac{d[A]}{dt}$$

90) **Explain Question:**
We are given that $t_{1/2}$ vs concentration graph is a straight horizontal line, and asked to find the order (n) and time (t) for 50% decomposition at a different concentration.

Concept:
This question is based on (reaction order and half-life dependence)

Solution:



From the graph it is clear that half life is independent of concentration, So it is 1st order and half-life also remains same, it doesn't depend on initial concentration.

n = 1,

$t'_{1/2} = 10\ \text{min}$

Final Answer: Option (2)

BIOLOGY

91) NCERT XI Pg # 2

92) NCERT (XI) ; Pg.# 8

93)

NCERT (XI) ; Page No. # 5, 6

94) NCERT (XI) ; Page No. # 3

95) NCERT (XI) ; Page No. # 15

96)

NCERT-XI, Pg # 8

97)

NCERT-XI, Pg # 4

98)

NCERT (XI) ; Page No. # 23

99) NCERT (XI) ; Page No. # 24

100)

NCERT (XI) ; Page No. # 6,7

101) NCERT XI, Pg. # 6

102) NCERT (XIth) Pg. # 4

103)

NCERT (XI) ; Page No. # 4

104)

NCERT (XI) ; Page No. # 6

105) NCERT XI, Pg. # 8

106)

NCERT XI, Pg # 8 ; Table : 1.1

107) NCERT (XI) ; Page No. # 11

108)

NCERT (XI) ; Page No. # 11

109)

NCERT (XI) ; Page No. # 90, 91

110)

NCERT (XI) ; Page No. # 13

111)

NCERT XI Page No. # 90

112)

NCERT XI Page No. # 13

113)

NCERT XI Page No. # 14

114)

NCERT XI Page No. # 17, 18

115) NCERT (XIth) Pg. # 13

116)

NCERT XI, Pg. # 18

117)

NCERT (XI) ; Page No. # 13

118) NCERT (XI) ; Page No. # 18, 26
The correct answer is **3. They all may be used as food or food supplement.**
Explanation: **Porphyra, Agaricus, Chlorella,** and **Spirulina** are all used as food or food supplements.
• **Porphyra** is a type of red algae used in certain cuisines.
• **Agaricus** is a genus of mushrooms, commonly known as edible mushrooms.
• **Chlorella(Non-motile green algae)** and **Spirulina(Blue green algae)** are consumed for their nutritional benefits, often as supplements.

119) NCERT-XI, Pg. # 15

120) NCERT PG. 7, 8

121)

NCERT (XI) ; Page No. # 17, 18

122)

NCERT (XI) ; Page No. # 13

123)

NCERT (XI) ; Page No. # 14, 18, 91, 32

124)

NCERT (XI) ; Page No. # 14, 15, 16

125)

NCERT (XI) ; Page No. # 14, 15

126)

NCERT (XI) ; Page No. # 11

127)

NCERT (XI) ; Page No. # 20, 21

128)

NCERT (XI) ; Page No. # 15, 18, 13

129)

NCERT (XI) ; Page No. # 15

130)

NCERT (XI) ; Page No. # 14, 15

131)

NCERT (XI) ; Page No. # 17

132) NCERT (XI) Page No. # 17

133)

NCERT (XI) Page No. # 17

134)

NCERT (XI) Page No. # 17

135) NCERT Pg. # 18

136) NCERT Pg no. 18

137) NCERT (XI) Pg # 16

138)

NCERT (XI) Page No. # 17, 18

139)

NCERT (XI) Page No. # 20, 21

140)

NCERT (XI) Page No. # 20, 21

141) NCERT-XIth, Pg. # 32

142) NCERT-XIth, Pg. # 30, 32

143) NCERT-XIth, Pg. # 32

144) NCERT-XIth, Pg. # 32

145) NCERT-XIth, Pg. # 27

146) NCERT-XIth, Pg. # 33

147) NCERT-XIth, Pg. # 29

148) NCERT-XIth, Pg. # 30

149) NCERT-XIth, Pg. # 30

150) NCERT-XIth, Pg. # 26

151) NCERT-XIth, Pg. # 32

152) NCERT-XIth, Pg. # 32, 33

153) NCERT-XIth, Pg. # 33

154) NCERT-XIth, Pg. # 32

155) NCERT-XIth, Pg. # 29, 33

156) NCERT-XIth, Pg. # 26

157) NCERT-XIth, Pg. # 35

158) NCERT-XIth, Pg. # 32

159)

NCERT-XI Pg#28,29,30,32

160) NCERT-XIth, Pg. # 32

161) NCERT-XIth, Pg. # 26,29,33

162) NCERT-XIth, Pg. # 26,30,32

163) NCERT-XIth, Pg. # 33

164) NCERT-XIth, Pg. # 32, 33

165) NCERT-XIth, Pg. # 25,29,32,33

166) NCERT-XIth, Pg. # 26

167) NCERT-XIth, Pg. # 32

168) NCERT-XIth, Pg. # 27

169) NCERT-XIth, Pg. # 29

170) NCERT-XIth, Pg. # 32

171) NCERT-XIth, Pg. # 27

172) NCERT-XIth, Pg. # 33

173) NCERT-XIth, Pg. # 33

174) NCERT-XIth, Pg. # 29

175) NCERT-XIth, Pg. # 29,32,33

176) NCERT-XIth, Pg. # 27

177) NCERT-XIth, Pg. # 29

178) NCERT-XIth, Pg. # 30

179) NCERT-XIth, Pg. # 32

180) NCERT-XIth, Pg. # 32